



An intelligent measurement framework for multi-type rock detection and classification in geological images

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ABSTRACT

Lithology identification is a core process in geological surveys, mineral exploration and engineering modeling. Traditional methods, relying on manual expertise and laboratory analysis, suffer from limitations such as low efficiency, high cost and strong subjectivity, making it challenging to meet the demand for automatic identification of multi-type rock targets in complex field environments. Addressing these challenges, this paper proposes a deep learning lithology detection model based on YOLOv12. We construct an object detection system targeting nine typical rocks across three major categories: igneous, metamorphic and sedimentary rocks and systematically optimize the model's performance bottlenecks in small object recognition, feature redundancy compression and boundary discrimination accuracy. Methodologically, we introduce an Area Attention (AA) mechanism and a Residual Efficient Layer Aggregation Network (R-ELAN) to enhance the perception ability of texture structures. Furthermore, we integrate a multi-scale image strategy and lightweight inference optimization during model training to adapt to the complex backgrounds and diverse compositions prevalent in geological images. In experimental evaluation, the proposed model achieved significant performance improvement on a standard test set, ultimately reaching 0.9300 and 0.8841 for mAP@50 and mAP@50–95, respectively. Precision, Recall and F1-score were 0.9566, 0.8817 and 0.9176, representing an improvement of over 4 times compared to the initial state. Regarding loss functions, Box_loss, Cls_loss and Dfl_loss all stably decreased to levels significantly lower than their initial values, validating the network structure's effective convergence and generalization ability in geological images. In typical application scenarios (e.g., uneven illumination, mineral occlusion, texture-background interference overlap), the model can stably perform lithology target identification and spatial localization, demonstrating good small object decoupling ability and texture adaptability. The results indicate that YOLOv12 possesses strong automatic identification capability under high-complexity geological images, providing key technical support for building intelligent and automated geological information sensing systems and exhibiting significant theoretical value and engineering application potential.

1. Introduction

Accurate identification of rock types is fundamental in geological science, providing crucial support for mineral resource surveys, structural analysis, geological hazard early warning and engineering geological assessment [1]. For instance, in mineral exploration, precise delineation of ore-hosting and surrounding rocks directly determines the accuracy of ore body modeling and resource estimation [2]; in geological hazard assessment, identifying weak rock layers and structural features is key to building reliable slope stability models [3]. Igneous,

metamorphic and sedimentary rocks, as the three basic rock types, reflect geological processes such as magmatic activity, tectonic metamorphism and sedimentary environment evolution, respectively, forming the basic units for Earth system research and natural resource development [4–7]. However, traditional rock identification methods, whether reliant on laboratory analysis or expert visual inspection in the field, exhibit significant limitations. Laboratory methods, such as microscopic thin-section analysis and X-ray diffraction (XRD), can provide precise mineralogical and structural information [8,9], but they are highly dependent on core sampling. Core sampling itself is constrained

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by the quantity and location of samples, and subsequent manual interpretation is subject to subjectivity, making a comprehensive and objective characterization of geological bodies highly challenging [10,11]. Concurrently, these analytical methods are often costly and time-consuming. In field settings, direct observation by experts is susceptible to interference from objective factors such as sample weathering, variable lighting conditions, and surface cover, which compromises the consistency and reliability of identification results [12–16]. These inherent bottlenecks render traditional methods inadequate to meet the urgent demands of modern geological surveys for large-scale, high-efficiency, and intelligent solutions [12–14,17].

In recent years, computer vision (CV) and deep learning (DL) technologies have achieved significant progress in geoscience image analysis, offering a new path for lithology identification. The research has evolved through several distinct tasks: image classification, semantic/instance segmentation, and object detection [12–14,18,19,20]. Early studies often employed handcrafted features such as color histograms and texture statistics combined with traditional machine learning methods (e.g., K-Nearest Neighbors, Support Vector Machines, Random Forests) for rock image classification [4,5,21,22]. While these methods showed some effectiveness in small-scale experiments, they suffered from high dependency on feature engineering, weak generalization ability and insufficient adaptability to complex field images. Subsequently, deep learning models, particularly Convolutional Neural Networks (CNNs), were widely introduced, significantly enhancing the capabilities of automatic feature extraction and diverse image recognition [12–14,23,24]. Existing research has demonstrated good performance of CNN structures in tasks such as overall rock image classification and pixel-level segmentation of microscopic core sections [25–30]. Furthermore, the field is rapidly advancing towards more powerful architectures like Vision Transformers (ViT) and hybrid CNN-Transformer models (e.g., MobileViT) to create fast and lightweight classifiers suitable for deployment on edge devices [31,32]. Researchers are also exploring novel training strategies, such as adaptive loss functions, to improve performance on challenging and imbalanced datasets [33].

Another significant research direction is semantic segmentation, which performs pixel-level classification. This is particularly valuable for detailed petrographic analysis of thin sections. Recent studies have successfully employed dual-modal Vision Transformers to segment sediment grains from both plane-polarized (PPL) and cross-polarized (XPL) light images [12–14]. Concurrently, state-of-the-art CNN-based models like YOLOv11 have been used for the semantic segmentation of specific minerals in thin sections, with a growing emphasis on model explainability to understand which visual features (e.g., shape, color, contrast) the model prioritizes [34].

However, both image classification and semantic segmentation have inherent limitations for field applications. Classification fails to localize individual rock instances, while segmentation is computationally intensive and primarily focused on microscopic analysis [35–38]. There is a critical need for a method that can simultaneously identify multiple rock instances and localize their spatial positions within a single, complex image, such as a corebox or a field outcrop. This task is best addressed by object detection.

Object Detection technology has recently emerged as a significant breakthrough in applying deep learning to image analysis. Unlike image classification and segmentation tasks, object detection performs the dual tasks of “category recognition + precise localization” for multiple objects within an image. This makes it particularly suitable for the intelligent recognition of images containing multiple lithological particles, mineral assemblages, or rock fragments. This research direction is gaining traction, with recent studies applying YOLO-based models for the simultaneous classification and localization of rock cores [39] and for instance segmentation of geological features like rock fractures [40]. Mainstream object detection frameworks are broadly categorized into two-stage (e.g., Faster R-CNN) and one-stage (e.g., YOLO series)

detectors. The former generally offers higher accuracy, while the latter provides advantages in speed and deployment cost, making it especially suitable for field or industrial scenarios with high demands for inference efficiency and device adaptability [41,42]. The YOLO (You Only Look Once) series, a prominent representative of one-stage detectors, has continuously evolved from YOLOv1 to YOLOv12, with steady improvements in detection accuracy and computational efficiency. It has demonstrated excellent performance in various fields, including object detection, instance segmentation, autonomous driving and remote sensing image analysis [43–45] and has gradually been extended to geoscience image tasks such as rock identification [46,47]. Models like YOLOv8-seg and YOLOv9-Seg have achieved higher detection accuracy and generalization ability in object segmentation tasks through improvements in Feature Pyramid Networks (FPN), enhanced attention mechanisms and optimized loss functions [48,49]. YOLOv10 further optimized the model structure by introducing more lightweight computational modules, making it more suitable for resource-constrained devices like edge computing and UAV platforms [50]. YOLOv11, which has already been applied to petrographic analysis [34] showed significant improvements in multi-scale object recognition and adaptability to complex backgrounds [51–53]. YOLOv12 achieves comprehensive improvements in inference efficiency, edge deployment adaptability and recognition robustness under low-quality images, making it an ideal choice for mineral identification [54,55].

Despite substantial progress in deep learning for rock image classification and semantic segmentation, research focusing on rock object detection, particularly on the simultaneous localization and identification of multiple independent igneous, metamorphic, and sedimentary rock instances within a single image, remains relatively scarce. Existing studies have yet to sufficiently explore the potential of efficient and high-precision one-stage detectors such as the YOLO family for unified detection across typical geological lithologies. To bridge this gap, this paper introduces a multi-type rock object detection approach based on YOLOv12, tailored for multi-object recognition and spatial localization in typical rock imagery. A representative multi-class dataset is constructed by including common igneous (basalt, diorite, granite), metamorphic (marble, quartzite), and sedimentary rocks (chalk, limestone, sandstone, coal). Model optimization and comprehensive performance evaluation are conducted based on YOLOv12 to verify detection accuracy, robustness, and inference efficiency under different lithological characteristics and complex imaging conditions. As illustrated in Fig. 1, the proposed method provides a technically feasible and scalable solution for automated geological image interpretation, facilitating efficient geological surveying, resource exploration, and engineering decision-making.

2. YOLOv12 model Introduction

2.1. Algorithm overview

YOLOv12 (released in 2025) represents the latest generation of high-performance one-stage object detectors. It further strengthens small-object detection accuracy, feature representation robustness, and deployment efficiency on resource-constrained platforms [56]. While retaining the YOLO family’s advantages in fast end-to-end inference and lightweight model design, YOLOv12 introduces several structural enhancements specifically beneficial for geoscience imagery, which often contains complex textures, heterogeneous backgrounds, and large variations in target scale. The main improvements include: (1) Area Attention (AA), which associates spatial localization with semantic cues to refine fine-grained feature perception and suppress redundant channel information, thereby improving computational efficiency. (2) Residual Efficient Layer Aggregation Network (R-ELAN), which enhances inter-layer semantic fusion and feature propagation, particularly aiding representation in texture-rich rock regions. (3) Large-Kernel Separable Convolution with a Position-Aware Module, expanding the

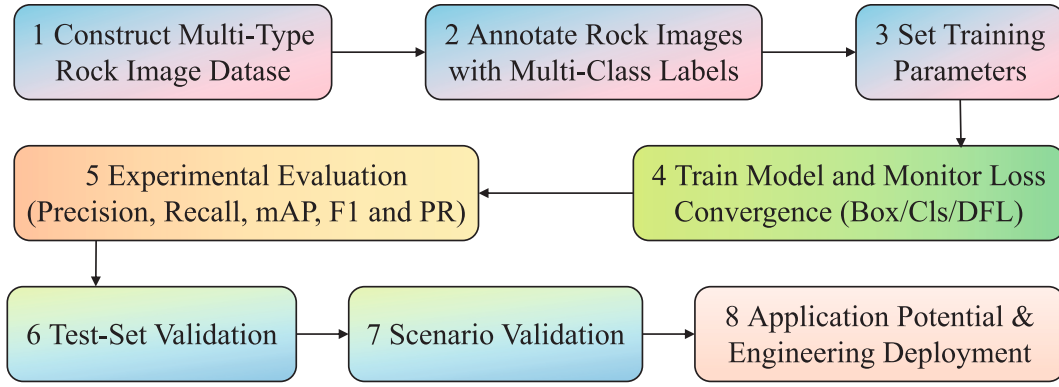


Fig. 1. Workflow of the proposed intelligent lithology detection and classification framework.

receptive field while preserving location-sensitive detail, thus improving recognition of geological structures such as cracks, joints, and particle arrangements. (4) FlashAttention, which reduces memory access overhead and accelerates attention computation, enabling high-throughput, low-latency inference on edge hardware commonly used in field geological applications. These improvements collectively make YOLOv12 well-suited for geological image detection tasks requiring robust recognition of fine-scale rock features under real-world resource constraints.

In terms of model scalability, YOLOv12 provides five variants with different computational complexities (N, S, M, L, X), enabling deployment across diverse scenarios ranging from lightweight edge devices to high-precision offline systems, thereby demonstrating strong portability and hardware adaptability [56]. Performance benchmarks on the MS COCO dataset further show that YOLOv12 achieves a more favorable balance between detection accuracy and computational cost. As illustrated in Fig. 2, YOLOv12 maintains superior mAP across various latency and FLOPs settings compared with YOLOv11, YOLOv10, and RT-DETR. Particularly, the lightweight YOLOv12-N variant reaches 40.6 % mAP on the NVIDIA T4 platform, exceeding YOLOv10-N and YOLOv11-N by 2.1 and 1.2 percentage points, respectively, indicating stronger detection capability under low-power constraints. Additionally, YOLOv12 natively supports unified modeling of both object detection and instance segmentation, enabling simultaneous generation of bounding boxes and pixel-level masks. This capability is especially beneficial for multi-granularity geological interpretation tasks such as

core image analysis and mixed lithology discrimination, extending the applicability of YOLO models toward integrated semantic analysis of rock structures and spatial relationships.

As shown in Table 1, a benchmark of seven YOLO variants (input size 640×640) on NVIDIA A100 and T4 GPUs was conducted to evaluate suitability for deployment in geological image measurement scenarios. Latency comparisons were based on intra-device trade-offs due to differences in GPU capabilities. On T4, where computational resources are representative of practical field or edge environments, YOLOv12x achieves the highest accuracy among the evaluated models (55.2 mAP), which is crucial for detecting fine rock features and subtle geological structures. Although the latency increases marginally compared with YOLOv10x (≈ 11.79 ms vs. 10.70 ms), the improvement in detection performance demonstrates a more favorable trade-off for precision-critical geological tasks. On A100, YOLOv9e yields strong

Table 1

Performance analysis of YOLO models for geological detection tasks.

Model	Size (pixels)	mAP ₅₀₋₉₅	TensorRT (ms)	Params	FLOPs
YOLOv5xu	640	53.2	A100: 3.81	97.2 M	246.4B
YOLOv7-x	640	53.1	A100: 3.69	71.3 M	189.9G
YOLOv8x	640	53.9	A100: 3.53	68.2 M	257.8B
YOLOv9e	640	55.6	A100: 3.41	58.1 M	192.5B
YOLOv10x	640	54.4	T4: 10.70	29.5 M	160.4G
YOLO11x	640	54.7	T4: 11.30	56.9 M	194.9B
YOLO12x	640	55.2	T4: 11.79	59.1 M	1990.0B

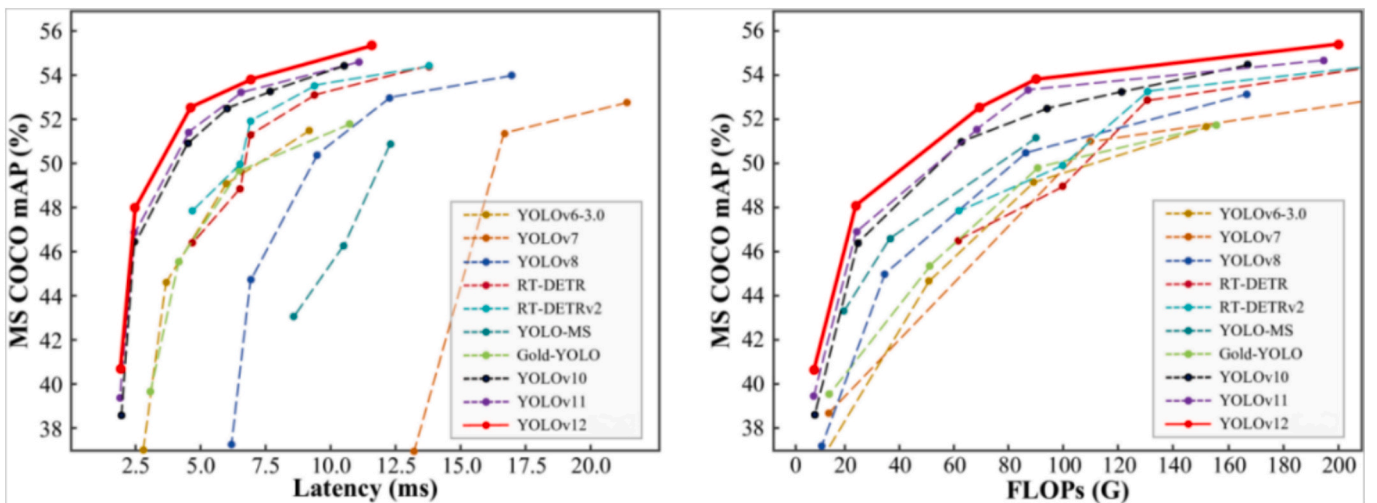


Fig. 2. Latency-accuracy and FLOPs-accuracy trade-offs across popular detection models [57]. (This figure compares the proposed YOLOv12 model against existing mainstream detectors in terms of real-time inference speed (latency vs. accuracy, left) and computational complexity (FLOPs vs. accuracy, right). The comparison demonstrates YOLOv12's superior balance between detection accuracy and deployment efficiency.)

accuracy–efficiency performance. However, as geological imaging in realistic deployments often relies on resource-limited endpoints, the T4-oriented evaluation holds greater practical relevance for the proposed work. These results highlight that YOLOv12x presents the optimal balance for geological image applications where detection accuracy is prioritized over minimal latency, especially in scenarios requiring reliable differentiation of multi-type rock materials (see Table 2).

In summary, YOLOv12 delivers significant advancements in detection accuracy, lightweight architectural design, multi-scale adaptability, and deployment efficiency, demonstrating strong engineering feasibility and application potential. Its capability for fast and reliable identification and localization of diverse rock types makes it particularly advantageous for geological surveying and resource exploration scenarios, where high-precision instance-level perception is required under real-time or resource-constrained conditions.

2.2. Overall network architecture

YOLOv12 represents a significant advancement in the YOLO series algorithms, particularly enhancing multi-scale object detection capabilities and feature representation accuracy [57]. Its overall architecture adheres to the conventional “Backbone–Neck–Head” three-stage structure. The Backbone is responsible for extracting fundamental image features, the Neck facilitates the fusion and enhancement of semantic features and the Head generates the final detection results. While preserving the inherent efficiency and lightweight nature of the YOLO series, this architecture incorporates several novel components: the Area Attention mechanism, implemented as adaptive attention modules (A2C2f); the Residual Efficient Layer Aggregation Network (R-ELAN), which utilizes improved residual structures (C3K2); and refined multi-scale fusion paths. These additions are specifically designed to address the challenges of object detection in complex geological images characterized by fine textures and substantial scale variations. Fig. 3 illustrates the detailed network structure of YOLOv12, where efficient inter-layer information exchange and feature enhancement are achieved through multi-level convolutions, upsampling and feature fusion across modules.

Within the Backbone, standard convolutional units (nodes 1, 2, 4, 6, 8) are employed to perform initial feature extraction. A hierarchical downsampling process is applied layer by layer, which progressively reduces the spatial resolution while increasing the number of channels, thereby enabling the network to extract high-level semantic representations. Two C3K2 modules (nodes 3, 5), which are implemented as lightweight residual blocks, are integrated into the intermediate stages in order to enhance the non-linear modeling capacity. The C3K2 module, which is derived from the CSP architecture used in YOLOv5, incorporates a refined channel interaction design that allows the network to more effectively and stably capture the relationships between local features and global context throughout the forward propagation.

The Backbone also includes two A2C2f modules (nodes 7, 9), which combine a self-attention mechanism with a multi-scale convolutional structure. These modules are designed to assign dynamic weights to feature responses across different spatial positions and channel dimensions, which enables the model to better encode regions that exhibit

heterogeneous textures or irregular geometries. This design contributes to improved generalization performance, especially for lithological categories that contain significant internal structural variations, such as igneous and metamorphic rocks.

The Neck component highlights YOLOv12’s capability to integrate multi-scale features in a hierarchical manner. Two upsampling operations (nodes 10, 13) and four feature concatenation steps (nodes 11, 14, 17, 20) are included, which facilitate the bottom-up aggregation of feature maps from different levels of the Backbone. A series of three A2C2f modules (nodes 12, 15, 18) are applied subsequently, which allow for deep semantic fusion across layers by preserving fine-grained information from early stages while incorporating abstract contextual representations from deeper layers. This is followed by additional convolutional layers (nodes 16, 19) and a final C3K2 module (node 21), which are used to further refine the fused features, particularly enhancing the model’s ability to recognize small-scale objects and identify regions where class boundaries are ambiguous. When compared to traditional feature pyramid networks such as FPN or PAN, the Neck design in YOLOv12 provides a more consistent and computationally efficient feature integration pathway, which is advantageous for detecting granular structures in high-resolution geological images.

The Head module adopts a three-branch structure (nodes 22–1, 22–2, 22–3), where each branch is connected to a feature map with a specific resolution. These branches are respectively responsible for detecting large-, medium- and small-scale targets. Each branch performs bounding box regression, class probability estimation, and confidence scoring, which ensures accurate detection across a wide range of object scales. This architecture is particularly effective in handling rock structures that display significant variation in size, including coarse mineral clusters, fine-grained aggregates, and complete lithological units. Moreover, the use of multiple resolution branches contributes to improved contextual awareness, which helps reduce both false positive and false negative predictions in geological images that contain occlusion, illumination variation, or complex background interference.

The overall architecture of the YOLOv12 model is composed of multiple key sub-modules with distinct functionalities and optimized structures, as illustrated in Fig. 4. These primarily include the basic convolutional unit (Conv), the C3K2 module, the A2C2f module, and the decoupled detection head module (Detect). Each of these modules serves a specific role in the object detection pipeline, collectively enhancing the model’s comprehensive performance in feature extraction, semantic modeling and detection accuracy. This architecture is particularly well-suited for recognition tasks in rock images characterized by complex textures and significant scale variations.

As depicted in Fig. 4(a), the Conv module is the fundamental building block within the YOLOv12 network. Its internal structure consists of a standard 2D convolution (Conv2d), Batch Normalization (BatchNorm2d) and a SiLU activation function. This module effectively mitigates feature distribution shift and enhances non-linear representation capabilities while maintaining a low computational footprint. It is extensively employed across various layers within the Backbone, Neck and Head structures.

Fig. 4(b) illustrates the C3K2 module, a core component for modeling intermediate-level semantics in YOLOv12, which features two structural

Table 2
Comparative overview of employed loss functions.

	Box_loss		Cls_loss		Dfl_loss	
	Train	Valid	Train	Valid	Train	Valid
Initial value	1.3274	1.3926	2.6408	2.8430	1.6857	1.6808
Maximum value	1.4261 (Epoch 4)	4.1660 (Epoch 2)	2.6408 (Epoch 1)	10.6031 (Epoch 4)	1.6857 (Epoch 1)	3.9430 (Epoch 2)
Terminal Value	0.1931	0.4616	0.1284	0.6506	0.8462	0.9410
Reduction rate	−85.45 %	−66.86 %	−95.14 %	−77.12 %	−49.80 %	−44.01 %
Reduction rate relative to maximum value	−86.46 %	−82.92 %	−95.14 %	−93.86 %	−49.80 %	−76.13 %

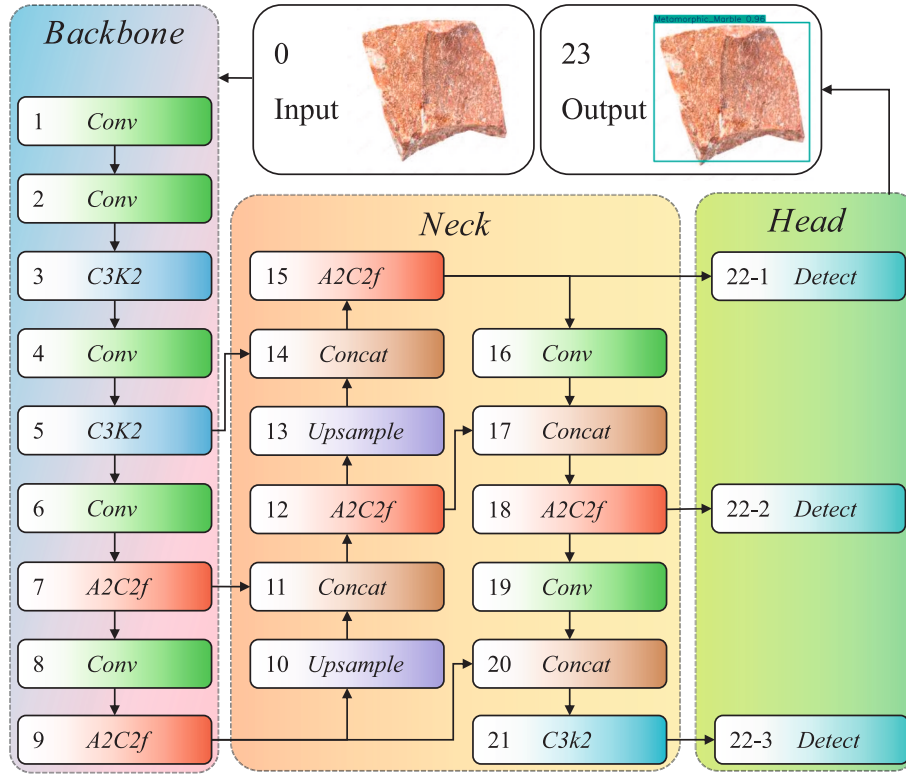


Fig. 3. Structural Overview of the YOLOv12 Network. (This figure illustrates the three-stage architecture of YOLOv12: Backbone, Neck, and Head. It highlights the flow of feature extraction, semantic enhancement, and detection output, serving as a structural reference for subsequent module-level analysis.)

configurations. When the parameter *c3k* is set to True, the module performs a Split operation on the input feature map, utilizes three parallel C3k structures to extract multi-channel features, integrates the outputs via a Concatenation operation, followed by convolutional fusion. Conversely, when *c3k* is False, the module reverts to a stacked configuration of typical Bottleneck structures. The C3k sub-module is composed of multiple Bottleneck residual blocks that internally maintain stable gradient flow via skip connections, thereby improving training efficiency. The C3K2 module is designed to enhance the discriminative power of intermediate semantic features while concurrently strengthening the model's perception of fine-grained information such as textures and structural boundaries, making it a critical supporting component for structural recognition in rock images.

As shown in Fig. 4(c), the A2C2f module further enhances the network's capacity to focus on salient feature regions. This module comprises two parallel Attention Blocks (ABlock) and a fusion convolutional module, with the final output generated by adding the result to the input features via a residual connection. Each ABlock incorporates an A2 attention mechanism to capture spatial saliency and leverages a Multi-Layer Perceptron (MLP) to model global contextual relationships, thereby achieving the dual functions of noise suppression and target feature enhancement. This module is repeatedly integrated throughout the Backbone and Neck of YOLOv12, significantly enhancing the model's responsiveness to tiny objects or texture transition regions under complex backgrounds.

Fig. 4(d) presents the Detect module, which adopts a decoupled design philosophy. It independently models bounding box regression and object classification tasks through two distinct sub-paths to minimize interference between the optimization processes of these tasks. The regression branch consists of a series of convolutional layers and is optimized using Box regression loss (Box Loss). The classification branch incorporates Depthwise Separable Grouped Convolution (DWG Conv) to reduce computational cost and utilizes Classification Loss (Cls Loss) to compute predicted class probabilities. This design renders the detection

head more flexible and scalable while balancing computational efficiency and accuracy requirements during inference.

Overall, the key structural innovations introduced in YOLOv12 include: (1) Repeated integration of A2C2f modules within the Backbone and Neck to enhance feature map representation quality and discriminative power via attention mechanisms. (2) The incorporation of the C3K2 module significantly optimizes the feature channel configuration within residual connections, contributing to improved modeling efficiency in intermediate layers. (3) Multi-level upsampling and concatenation strategies establish an effective bridge between deep semantic information and shallow textural details. (4) A three-branch detection head structure enhances the model's comprehensive responsiveness to multi-scale targets. Collectively, these design choices constitute a deep object detection framework characterized by a well-structured architecture, high computational efficiency and robust detection accuracy. It is particularly well-suited for recognizing multi-type targets with complex structures and significant scale variations in rock images, demonstrating high potential for deployment and significant research value in applications such as intelligent geological analysis and digital core interpretation.

3. Dataset construction and model training

3.1. Dataset construction

To support research on object detection algorithms for multi-type rock images, this study compiled an image dataset comprising typical rock categories. The dataset was primarily sourced from the Rock Classification Dataset [58], and it includes three major rock classes: Igneous, Metamorphic, and Sedimentary rocks. A total of nine detailed lithology types were selected: Basalt, Diorite, and Granite within the igneous rock category; Marble and Quartzite within the metamorphic category; and Chalk, Coal, Limestone, and Sandstone within the sedimentary category.

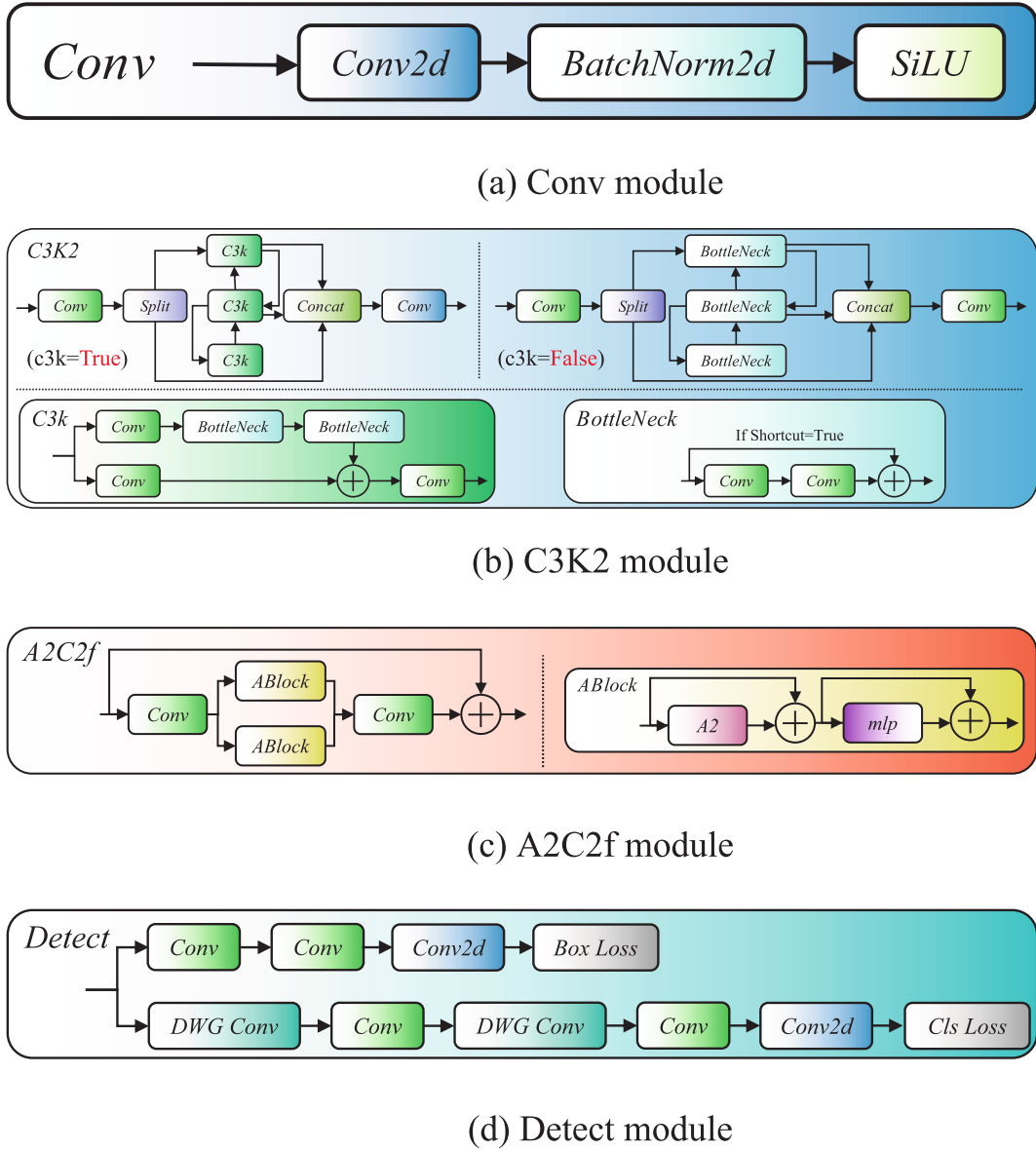


Fig. 4. Key Modules Constituting the YOLOv12 Framework [57]. (Subfigures (a)–(d) depict the core architectural units in YOLOv12: (a) Standard convolutional unit; (b) C3K2 module: a lightweight residual block for enhancing mid-level feature representation; (c) A2C2f module: an attention-augmented convolutional unit for improving spatial-channel feature sensitivity; (d) detect module: responsible for outputting final predictions across multi-scale feature maps.)

The dataset was constructed with careful consideration of complexity levels across data subsets. Specifically, the training and validation sets consist of images with relatively clean and homogeneous backgrounds, which help stabilize the learning process and minimize noise during optimization. In contrast, the test set was intentionally designed to contain samples with higher scene complexity, including variable lighting conditions, multi-lithology overlaps, and dense microstructures, in order to evaluate the model's generalization performance under realistic geological scenarios. It should be noted that the quantitative and visual analysis of these complex cases is further explored in Section 5: Application Case Analysis, as it focuses on application-level interpretability beyond the scope of dataset construction.

3.2. Data annotation

Annotation follows the YOLO format, where each object is recorded as a single line of text with five fields: `<class_id>` `<x_center>`

`<y_center>` `<width>` `<height>`. `<class_id>` represents the class index, while the latter four fields denote the x and y coordinates of the bounding box center, width, and height, respectively, all normalized to the image dimensions (ranging from 0 to 1). As illustrated in Fig. 5, the annotation '0 0.492788 0.508413 0.836538 0.78125' indicates an object with class ID 0, centered at 49.28 % of the image width and 50.84 % of the height, with a bounding box width of 83.65 % and a height of 78.13 %. All annotations were manually generated using Labelme [59], an open-source image annotation tool that supports polygon and bounding-box labeling. After annotation, the outputs were converted into YOLO format through a custom script to enable seamless integration into the training pipeline.

Prior to inputting images into the model, they were uniformly scaled to 640×640 pixels. Preprocessing operations such as brightness normalization, color enhancement, and edge sharpening were also applied. Furthermore, during the training phase, various data augmentation strategies were employed to enhance the model's robustness against variations in scale, texture, and imaging conditions. These

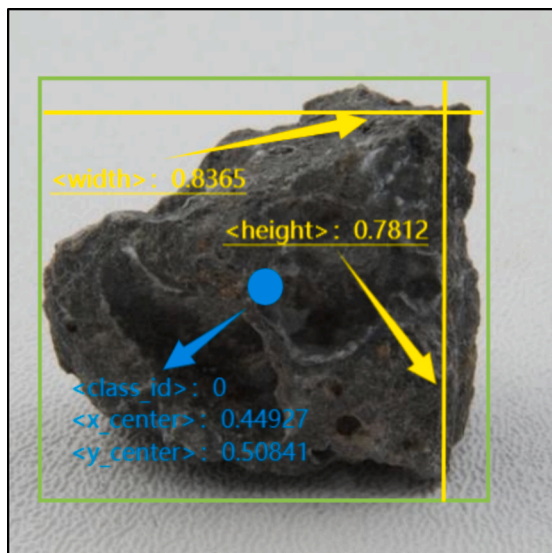


Fig. 5. Illustration of bounding box annotation.

strategies included rotation, flipping, color jitter, blurring, CutMix, and MixUp.

As shown in Fig. 6, the dataset comprises a total of 6282 images, divided into a training set of 3787 images, a validation set of 991 images, and a test set of 504 images. The split was performed to ensure a balanced distribution of rock categories across each subset. Notably, the test set specifically includes image samples with challenging characteristics such as complex illumination, impure backgrounds, or blurry structural boundaries, designed to evaluate the model's adaptive performance under non-ideal imaging conditions. This dataset exhibits considerable diversity in terms of rock category composition, image texture variations, and structural scale coverage, thereby providing fundamental data support for research on object detection models targeting igneous, metamorphic, and sedimentary rock images.

3.3. Model training parameters and strategies

For the multi-type rock image object detection task in this study, the following training parameters and strategies were adopted to enable the YOLOv12 model to effectively learn the structural features and texture differences present in igneous, metamorphic, and sedimentary rock images:



Fig. 6. Representative samples from the dataset.

- (1) Epochs: The training duration was set to 1000 epochs to facilitate sufficient model fitting for multi-scale targets and complex structures across different lithology categories. In practice, the model reached convergence criteria at epoch 919, indicated by the stabilization of validation loss, and training was automatically terminated early, thereby preventing overfitting and reducing unnecessary computational costs.
- (2) Batch Size: A batch size of 128 was selected. This size improves gradient estimation stability while ensuring GPU memory efficiency and accelerates overall convergence.
- (3) Input Image Size (Imgsz): The input image resolution was set to 1280×1280 pixels to enhance the model's responsiveness to detailed regions within rock images. This resolution is particularly suitable for tasks such as localizing crystalline grains in igneous rocks, identifying edges of banded structures in metamorphic rocks, and detecting fine sedimentary layering in sedimentary rocks.
- (4) Pre-trained Weights: Pre-trained weights from YOLOv12x.pt were loaded. This model, pre-trained on large-scale general image datasets, possesses strong low-level feature extraction capabilities, which improves transfer learning efficiency and initial detection performance on rock image data.
- (5) Learning Rate: The initial learning rate was set to 0.0001 and dynamically adjusted using a Cosine Annealing Scheduler. A larger learning rate was maintained in the early training stage to promote rapid feature extraction, while gradually decreasing it in later stages to enhance parameter stability and model convergence quality.
- (6) Optimizer: The AdamW optimizer was employed. Its adaptive update mechanism and built-in weight decay strategy offer superior training stability and generalization capability compared to SGD, particularly when addressing challenges posed by complex textures and blurry boundaries in rock images.
- (7) Regularization: L2 regularization was incorporated to suppress weight inflation, further control model capacity, and reduce the risk of overfitting in regions with similar inter-lithology boundaries.

To ensure the reproducibility of all experiments, training was conducted in a controlled environment. The experiments were executed on a workstation equipped with three NVIDIA GeForce RTX 2080 Ti GPUs (each with 22 GB memory), an Intel i7-10700KF CPU and 128 GB RAM. The implementation was based on PyTorch 2.1.2 with CUDA 12.1 and cuDNN 8.9 support. The operating system was Ubuntu 20.04 LTS, and all training processes were executed in a Conda-managed virtual environment running Python 3.11. Dependency management was handled using Conda to ensure consistency across runs. Under full GPU utilization, the complete training procedure required approximately 240 h.

This comprehensive parameter configuration, combined with controlled augmentation strategies and a rigorously defined computing environment, was tailored to the texture heterogeneity, scale diversity, and boundary uncertainty inherent in rock image detection tasks, providing a robust foundation for accurate lithology recognition across igneous, metamorphic, and sedimentary rocks.

4. Analysis of training results

4.1. Evaluation and analysis of loss functions

The YOLOv12 model is optimized using a combination of several loss functions during the object detection process. These primarily include:

- (1) Box_Loss: This loss quantifies the positional error between predicted bounding boxes and ground truth bounding boxes. It commonly utilizes CIoU (Complete Intersection over Union) as the regression metric, which accounts for factors such as the

Intersection over Union, the distance between center points, and the consistency of aspect ratios between predicted and ground truth boxes. Box_Loss is a critical component for achieving high-precision object localization, as shown in Eq. (1).

- (2) Cls_Loss: Designed to supervise the model's capability for target class discrimination, Cls_Loss is typically formulated based on the cross-entropy loss function. In YOLOv12, it can be adapted using weighted cross-entropy or Focal Loss to address class imbalance issues. This loss term directly influences the model's ability to differentiate between multi-class lithology targets (e.g., igneous, metamorphic, sedimentary rocks), as shown in Eq. (2).
- (3) Dfl_Loss: Also known as Distribution Focal Loss (DFL), this loss is employed for achieving high-precision fitting of bounding box coordinates. DFL transforms continuous value regression into discrete distribution modeling. By learning the probability distribution of boundary locations, it enhances the model's fine-grained representation capability for target boundaries. This is particularly beneficial for detection tasks in rock images characterized by blurry structural boundaries and significant scale variations, as shown in Eq. (3).

Collectively, these three losses constitute the total loss function system of YOLOv12. They operate on the distinct sub-tasks of object perception, class discrimination and position regression, providing the fundamental loss function framework for object detection modeling in multi-type rock images

$$\text{Box_loss} = 1 - \text{CIoU} = 1 - \left(\text{IoU} - \frac{\rho^2(b_p, b_{gt})}{c^2} - \alpha v \right)$$

$$v = \frac{4}{\pi^2} \left(\arctan \frac{w_{gt}}{h_{gt}} - \arctan \frac{w}{h} \right)^2$$

$$\alpha = \frac{v}{(1 - \text{IoU}) + v} \quad (1)$$

[60] where IoU : Intersection over Union of the predicted box and the ground truth box; (b, b_{gt}) : Euclidean distance between the center points of the two boxes; c : diagonal length of the smallest enclosing box covering both the predicted and ground truth boxes; v : aspect ratio consistency measure; α : adaptive adjustment coefficient.

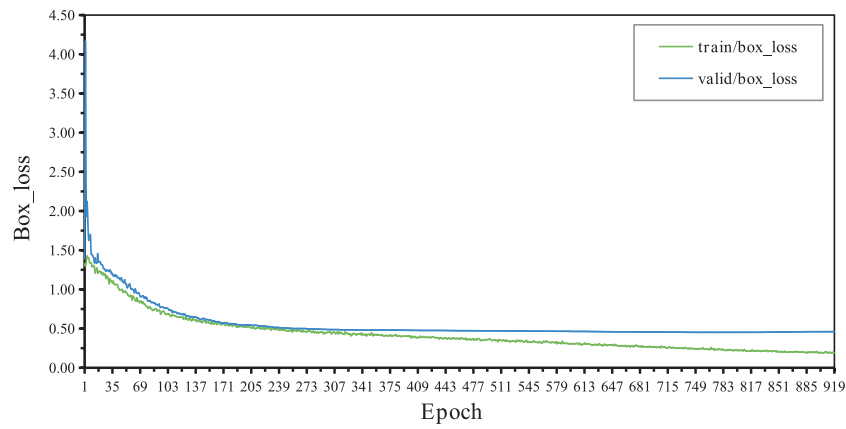
$$\text{Cls_loss} = - \sum_{i=1}^C [y_i \times \log(p_i)] \quad (2)$$

[60] where C : Total number of classes (9 classes in this study); y_i : True label for the i -th class; p_i : Predicted probability for the i -th class.

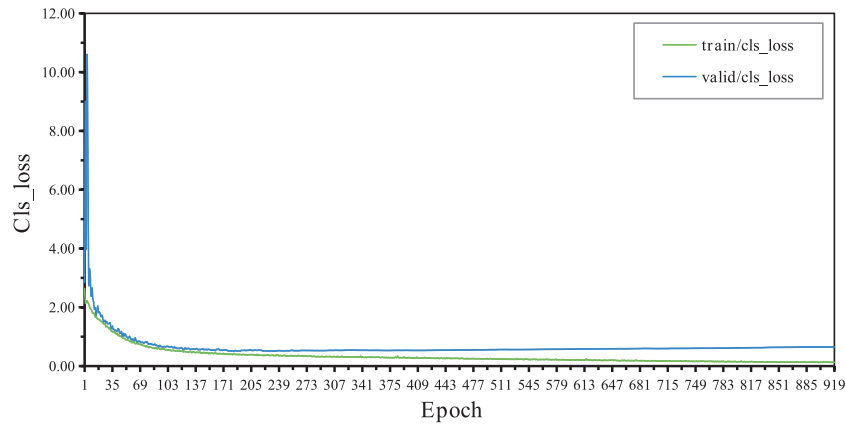
$$\text{Dfl_loss} = \sum_{j=1}^4 \sum_{i=0}^{K-1} q_{ij} \times (-\log p_{ij}) \quad (3)$$

[60] where $j \in \{l, t, r, b\}$: representing the four boundary regression directions (left, top, right, bottom); K : number of discrete location bins (e.g., $K = 16$); q_{ij} : the target distribution of the ground truth value projected onto the i -th bin in the j -th direction (typically assigned using adjacent weights); p_{ij} : the model's predicted softmax probability for the i -th bin in the j -th direction.

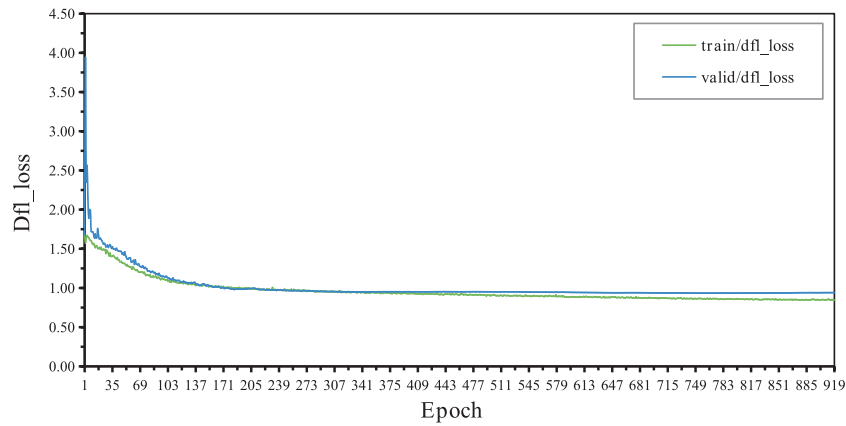
As illustrated in Fig. 7, the evolution of loss functions during the training of the YOLOv12 model for multi-type rock image object detection effectively reflects its learning capabilities and convergence behavior across different detection sub-tasks. Based on the trends observed in the curves of the three core loss functions (Box_Loss, Cls_Loss and Dfl_Loss) shown in Fig. 7, combined with the numerical statistics presented in Table 1, the following common characteristics can be summarized:



(a) Box_loss



(b) Cls_loss



(c) Dfl_loss

Fig. 7. Training and validation loss curves over epochs. (This figure tracks the evolution of Box_Loss, Cls_Loss, and Dfl_Loss for both training and validation sets during model training. The convergence behavior reflects the model's learning stability and optimization quality across localization, classification, and regression sub-tasks.)

- (1) All three loss functions are initially at relatively high values and decrease rapidly within the first 50 epochs, indicating the model's strong feature extraction capability in the early training phase.
- (2) Both the training and validation losses exhibit a clear convergence trend. Although the validation loss is consistently higher than the training loss, the gap between them remains small, with

- no significant oscillation or divergence observed. This indicates a stable training process and the absence of overfitting.
- (3) All three losses peak within the first 1–4 epochs, followed by a continuous decrease and subsequent stabilization. This suggests that after initial adaptive adjustments of key parameters, the model enters an effective optimization phase.

Specifically, as detailed in Table 1, Box_Loss reflects the model's capability for spatial fitting of object boundaries in rock images. On the training set, it decreased from 1.3274 to 0.1931, and on the validation set, from 1.3926 to 0.4616. The peak values were observed at Epoch 4 (train) and Epoch 2 (validation), reaching 1.4261 and 4.1660, respectively. The reduction rates were −85.45 % (train) and −66.86 % (validation), while the relative reduction rates with respect to the maximum value reached −86.46 % and −88.92 %, respectively. This significant decrease indicates YOLOv12's strong localization convergence ability across various lithological structures. It demonstrates the model's capacity to rapidly identify object edge regions such as grain boundaries, bedding interfaces and fracture contours in rock images, while maintaining low localization error. Furthermore, this loss function exhibits good robustness in identifying large grain boundaries in igneous rocks and maintains high consistency when dealing with bedding textures in sedimentary rocks. Its adaptability to objects with significant scale variations highlights YOLOv12's modeling capability for multi-scale boundary learning.

Cls_Loss, serving as a metric for multi-class classification error, decreased from 2.6408 to 0.1284 on the training set and from 2.8430 to 0.6506 on the validation set, corresponding to reduction rates of −95.14 % (train) and −77.12 % (validation). Notably, its peak value reached 10.6031 at Epoch 4 on the validation set, and the relative reduction rate from maximum was −93.86 %. This outcome demonstrates a significant enhancement in the model's ability to differentiate rock categories, even under complex backgrounds. Notably, it maintains low classification error even when texture overlap exists among multiple lithology types (e.g., granite-quartzite, limestone-chalk), exhibiting strong inter-class discriminability. Furthermore, the model maintains stable classification accuracy for samples in sedimentary rock images exhibiting uneven feature distribution or pore structures. This further substantiates YOLOv12's adaptability to non-ideal image structures.

Dfl_Loss, a crucial loss term for fine localization in position regression, decreased from 1.6857 to 0.8462 on the training set and from 1.6808 to 0.9410 on the validation set. The respective peak values were 1.6857 (Epoch 1, train) and 3.9430 (Epoch 2, valid), and the relative reduction rates were −49.80 % (train) and −76.13 % (validation). While its decrease rate is slower compared to Box_Loss and Cls_Loss, it exhibited good stability and a consistent convergence trend. This indicates that the model can still achieve relatively precise coordinate prediction through distributed position modeling when encountering complex regions in rock images characterized by blurry boundaries, continuous particles, or subtle texture transitions (e.g., gradual foliation zones in metamorphic rocks, boundaries of sandstone grain clusters). Particularly in lithological transition zones or mixed rock areas, the steady optimization of Dfl_Loss provides the model with the capability for micro-scale boundary representation, thereby enhancing overall object detection accuracy.

A deeper analysis reveals that the optimization of Box_Loss, Cls_Loss, and Dfl_Loss is not independent, but exhibits an intrinsic synergy that enhances model convergence and robustness. In the early stages of training, Cls_Loss and Box_Loss dominate the gradient updates, enabling the model to quickly establish class discrimination and coarse spatial localization. As training progresses, Dfl_Loss gradually assumes a more critical role by refining coordinate regression, especially in scenarios with ambiguous or fuzzy boundaries. This phase-wise shift in gradient dominance reduces the risk of conflicting updates and allows each loss function to specialize in different sub-tasks. Functionally, Cls_Loss focuses on inter-class separability, Box_Loss ensures global box alignment, and Dfl_Loss captures sub-pixel level localization through continuous distribution modeling. These three components form a complementary multi-objective supervision mechanism that balances semantic learning and geometric accuracy. In structurally complex scenarios such as marble fracture zones, where category ambiguity, texture continuity, and weak edges co-occur, this synergy becomes particularly critical: Cls_Loss helps maintain label consistency, Box_Loss anchors

approximate object regions, and Dfl_Loss progressively corrects boundary uncertainty. Moreover, Dfl_Loss acts as a stabilizing factor by continuing to provide meaningful gradients in areas where Box_Loss saturates, thereby extending the effective optimization window. Overall, this collaborative behavior across temporal phases, task dimensions, and spatial complexity levels enables YOLOv12 to achieve both rapid convergence and high-precision detection in challenging geological image environments.

4.2. Evaluation and analysis of performance metrics

To quantitatively evaluate the performance of the YOLOv12 model during training, five standard object detection metrics were adopted: Precision, Recall, F1 Score, mAP@50, and mAP@50–95.

As defined in Eq. (4), Precision measures the proportion of correctly identified targets among all predicted targets, reflecting the model's ability to minimize false positives. This is particularly important for igneous rock images, where fine-grained matrices and impurities can be mistaken for crystal grains.

Recall (Eq. (5)) evaluates the model's capacity to detect all actual targets, helping assess false negatives. It is critical for recognizing subtle or gradient-based features in metamorphic rocks, such as banding or microstructures.

The F1 Score (Eq. (6)), as the harmonic mean of Precision and Recall, offers a balanced assessment and is especially suitable for detecting lithological transition zones or mixed rock regions.

mAP@50 (Eq. (7)) evaluates average detection precision at an IoU threshold of 0.5, focusing on the model's ability to localize approximate target regions—such as mineral boundaries or crystalline zones in igneous and sedimentary rocks.

mAP@50–95 (Eq. (8)) is a stricter and more comprehensive indicator that averages precision across multiple IoU thresholds from 0.5 to 0.95. It captures boundary fitting performance and robustness, which are essential in complex cases involving blurry interfaces, multi-scale particles, or transitional lithologies.

$$\text{Precision} = \frac{TP}{TP + FP} \quad (4)$$

[61]

$$\text{Recall} = \frac{TP}{TP + FN} \quad (5)$$

[61]

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (6)$$

[61]

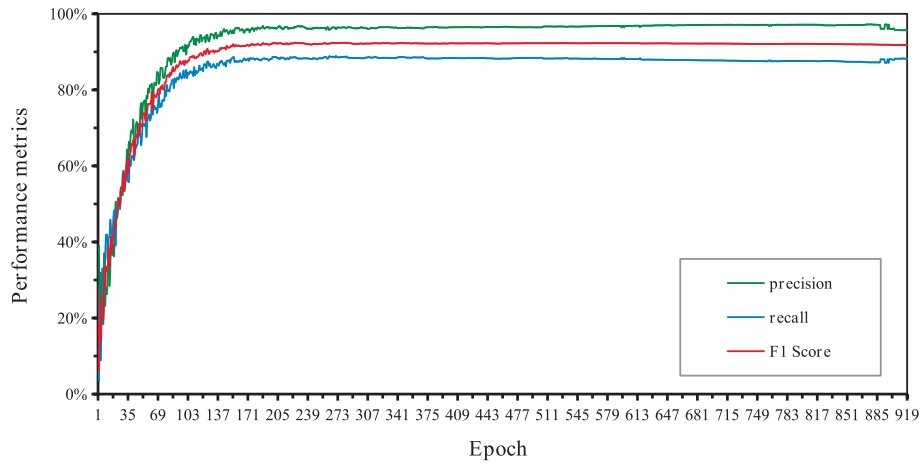
$$mAP_{50} = \frac{1}{N} \sum_{i=1}^N AP_{50,i} \quad (7)$$

[61]

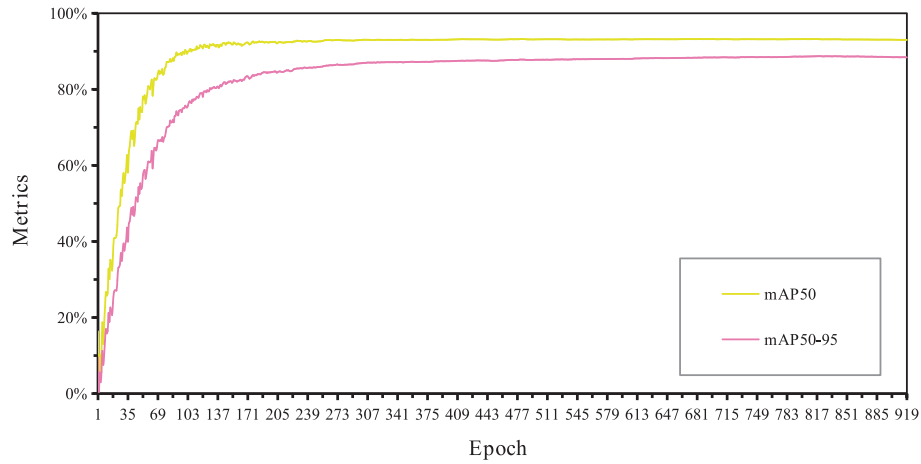
$$mAP_{50-95} = \frac{1}{N} \sum_{i=1}^N \frac{1}{K} \sum_{j=1}^K AP_{i,j} \quad (8)$$

[61] where TP (True Positive): Positive instances correctly detected by the model (correctly identified targets); FP (False Positive): Negative instances incorrectly detected as positive by the model (false detections); FN (False Negative): Positive instances not detected by the model (missed detections); N : Total number of target classes; $AP_{50,i}$: Average Precision for class i at an IoU threshold of 0.5; K : Number of different IoU thresholds (10 values from 0.5 to 0.95); $AP_{i,j}$: Average Precision for class i at the j -th IoU threshold.

Concerning object detection accuracy metrics, Fig. 8(b) and Table 3 illustrate that mAP50 and mAP50–95 reflect the model's boundary prediction capability under lenient and strict IoU thresholds,



(a) Performance Metrics



(b) Metrics

Fig. 8. Performance Metrics of YOLOv12 Throughout the Training Phase. (The plot illustrates the model's performance trends in terms of mAP@50, mAP@50–95, Precision, Recall, and F1 score. These curves provide insight into the detection effectiveness and model generalization capacity throughout training epochs.)

Table 3
Performance metrics of YOLOv12 throughout the training phase.

	Precision	Recall	F1 Score	mAP50	mAP50-95
Initial value	0.3791	0.2154	0.2747	0.1637	0.0958
Maximum value	0.1562 (Epoch 3)	0.0339 (Epoch 2)	0.0624 (Epoch 2)	0.0581 (Epoch 4)	0.0042 (Epoch 2)
Terminal value	0.9566	0.8817	0.9176	0.9300	0.8841
Growth rate	152.34%	309.34%	234.04%	468.26%	822.83%
Decrease rate relative to maximum value	512.58%	2500.06%	1370.84%	1501.43%	20899.29%

respectively. mAP50 rapidly increased from an initial value of 0.1637 to 0.9300, representing a growth rate of 468.26 %. Compared to its minimum value, it increased by 1501.43 %. This indicates that the model can accurately capture the approximate location of targets in images, making it suitable for targets with clear boundaries and strong textures, such as phenocrysts in igneous rocks and distinct bedding in sedimentary rocks. mAP50–95, serving as a stricter evaluation standard, measures the model's boundary fitting accuracy across multiple IoU thresholds. Its value increased from 0.0958 to 0.8841, an 822.83 % growth. Relative to its minimum value, it increased by an impressive 20899.29 %. This performance demonstrates that the model maintains high boundary

consistency and robustness even when dealing with image regions characterized by blurry boundaries, varying scales, or strong particle continuity (e.g., gradual foliation zones in metamorphic rocks or fine particle aggregation areas in sandstone). In summary, the YOLOv12 model demonstrated significant improvement and stable performance across multiple aspects, including classification accuracy, recall capability, prediction balance and boundary precision, in the multi-type rock image object detection task defined in this study. This highlights its high adaptability and reliability for target recognition and localization in complex lithology images, providing solid detection accuracy support for subsequent automated

lithology identification, rock structure analysis and intelligent structural interpretation.

As shown in Table 4 and Fig. 9, Aligned with the thematic goals of the intelligent measurement framework, the bootstrap-based intervals deliver multiple advantages. Methodologically, they exhibit strong robustness and generality, adapting well to the skewness and heterogeneity of geological images and reducing reliance on normal approximations or large-sample conditions—thereby enhancing credibility and transferability. Operationally, they are highly actionable: the lower bound in Eq. (13) can be adopted directly as a release threshold or acceptance criterion, maintaining a dependable “worst acceptable performance” under complex field conditions and adding a tangible layer of safety control. For model improvement, the gap between mAP_{50} and mAP_{50-95} spotlights high- IoU localization as a principal bottleneck; incorporating IoU -aware losses ($GIoU/DIoU/CIoU$), high-resolution multi-scale training, soft-NMS, and box refinement is expected to raise the interval’s lower bound and narrow its width upon regression testing via Eq. (11). Furthermore, the pattern of Precision exceeding Recall suggests conservative decision thresholds; cost-sensitive threshold learning and hard-example rebalancing can improve Recall without undue sacrifice of Precision, thereby better covering small, low-contrast, and occluded rock instances. Overall, the 95 % bootstrap confidence interval furnishes a rigorous, transparent, and reusable statistical foundation for the proposed framework: it is interpretable at the measurement layer, diagnostic at the analysis layer, and deployable at the decision layer. The approach not only upholds scientific rigor and reporting norms but also strengthens real-world reliability and industrial adaptability, offering clear and positive momentum for intelligent measurement and trustworthy evaluation in multi-type rock detection and classification.

4.3. Performance visualization based on PR curves

To further evaluate the object detection performance of YOLOv12 across different lithological categories, this study plots the Precision–Recall (P-R) curves for each rock type, as shown in Fig. 10. This figure illustrates the trade-off between precision and recall for each specific sub-category and annotates the average detection precision ($mAP@0.5$) for each class. Overall, the P-R curves for all categories consistently exhibit high precision and recall, reflecting the robust overall performance of YOLOv12 in multi-type rock image detection tasks.

Examining the performance across different rock categories, within the igneous rock category, Igneous Diorite (Diorite) achieved the highest $mAP@0.5$ of 0.972, indicating the model’s exceptional boundary discrimination and target consistency for this lithology. The detection performance for Igneous Basalt and Igneous Granite also remained high, with $mAP@0.5$ values of 0.951 and 0.908, respectively. Although Granite’s performance is slightly lower, it still maintains a strong overall result, demonstrating the model’s ability to reliably identify igneous

Table 4
Detection and localization metrics (precision, recall, F1, mAP_{50} , mAP_{50-95}) with 95% bootstrap confidence intervals.

	Precision	Recall	F1 Score	mAP_{50}	mAP_{50-95}
Mean	0.9332	0.8547	0.8917	0.9032	0.8366
Mean 95% CI lower bound	0.9254	0.8469	0.8839	0.8932	0.8214
Mean 95% CI upper bound	0.9406	0.8631	0.8993	0.9154	0.8572
Median	0.9649	0.8804	0.9217	0.9309	0.8761
Median 95% CI lower bound	0.9647	0.8795	0.9215	0.9308	0.8754
Median 95% CI upper bound	0.9653	0.8811	0.9218	0.9310	0.8778

rock structures with varying compositions and grain sizes.

In the metamorphic rock category, Marble and Quartzite achieved $mAP@0.5$ values of 0.950 and 0.873, respectively. Overall detection performance for this category is slightly lower than that for igneous rocks. The P-R curve for Quartzite shows a relatively faster drop, potentially attributable to the less distinct grain boundaries, smooth textures, and strong structural repeatability present in its images, suggesting a higher inherent discrimination difficulty for this rock type at the image level.

Within the sedimentary rock category, Coal and Limestone both achieved a high $mAP@0.5$ of 0.960, demonstrating the model’s strong perception capability for high-contrast features in these rock types (e.g., dark masses in coal, grain boundaries in limestone). Chalk and Sandstone achieved $mAP@0.5$ values of 0.880 and 0.912, respectively. While their performance is slightly lower, their P-R curves still remain close to the top-right corner, indicating the model’s stable detection capability even in images with relatively fine textures and uniform particle distribution.

Furthermore, the model’s average performance across all categories reached 0.930 $mAP@0.5$. The overall P-R curve exhibits a wide plateau, suggesting that YOLOv12 maintains high detection precision and consistency when encountering image samples with diverse lithologies, texture variations, and varied boundary morphologies, resulting in excellent overall object detection quality. The visualization provided by the P-R curves clearly illustrates the detection performance variations and potential challenges encountered by the model across different lithological targets, offering valuable insights for subsequent model refinement, data augmentation, and class balancing strategies.

4.4. Overall evaluation of detection accuracy and spatial distribution in multi-type rock images

4.4.1. Confusion matrix

To provide a fine-grained understanding of the model’s classification performance across multiple lithology categories, a confusion matrix is employed, as shown in Fig. 11(a). This matrix is a two-dimensional table where each row represents the predicted labels and each column corresponds to the actual (ground truth) labels. Diagonal entries indicate correctly classified instances, while off-diagonal values reflect misclassifications. The more concentrated the values along the diagonal, the better the overall classification accuracy of the model. Therefore, the confusion matrix is not only a standard tool for performance evaluation in multi-class classification tasks but also a highly interpretable diagnostic tool for identifying category-specific strengths and weaknesses.

As seen in Fig. 9a, the model demonstrates highly reliable detection across most rock types. Significant diagonal dominance can be observed, especially in the classes Sedimentary Coal (810), Sedimentary Limestone (641), Metamorphic Quartzite (400), Metamorphic Marble (373), and Sedimentary Sandstone (417), which confirms that the proposed framework can accurately differentiate major geological categories. This is a strong indicator of the model’s ability to capture lithology-specific textures, such as laminar bedding in sedimentary rocks, crystalline structures in metamorphic rocks, and grain homogeneity in coal deposits. Moreover, the model performs well even under class imbalance conditions, as indicated by consistent performance across both low-frequency (e.g., Basalt, Granite) and high-frequency categories (e.g., Coal, Sandstone). This suggests that the training strategy, including data augmentation and loss function optimization, effectively enhances model generalization. Additionally, the framework shows robustness in detecting classes with fine-scale textures or fuzzy boundaries, where traditional methods would often fail due to poor contrast or inter-class overlap.

Nonetheless, some confusion arises between visually or structurally similar rock types. For instance, Igneous Granite is misclassified as Metamorphic Marble in 96 instances, likely due to similar coarse-grained textures or transitional mineralogy. Other minor confusions

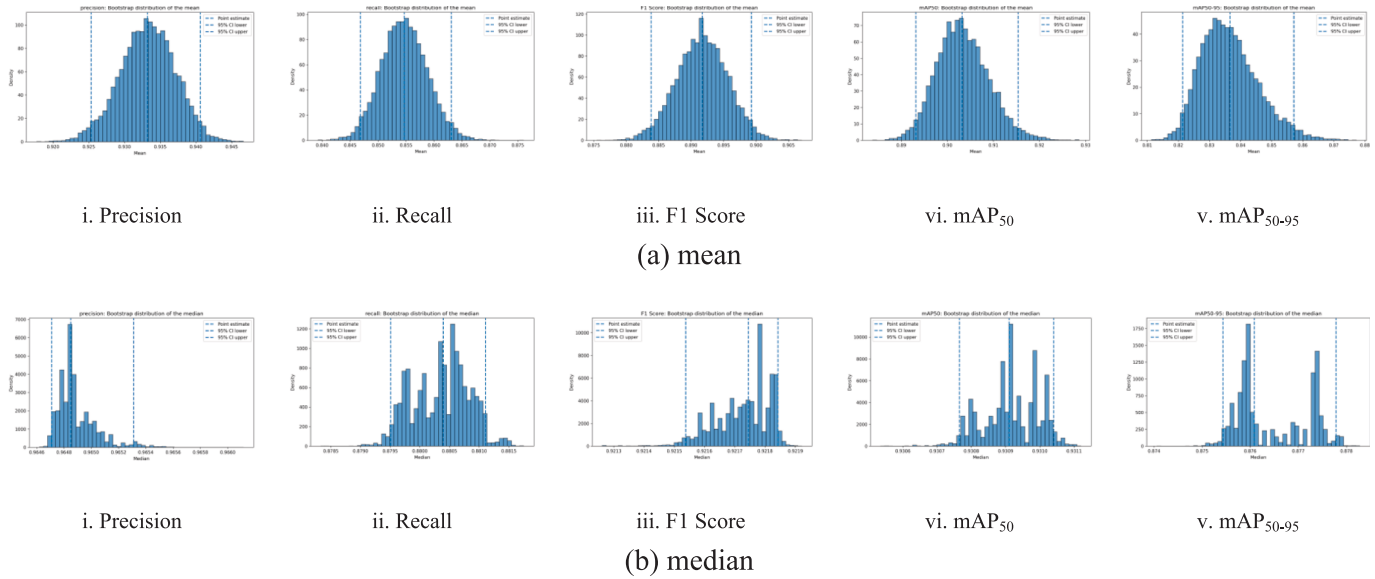


Fig. 9. Bootstrap distribution of the mean and median.

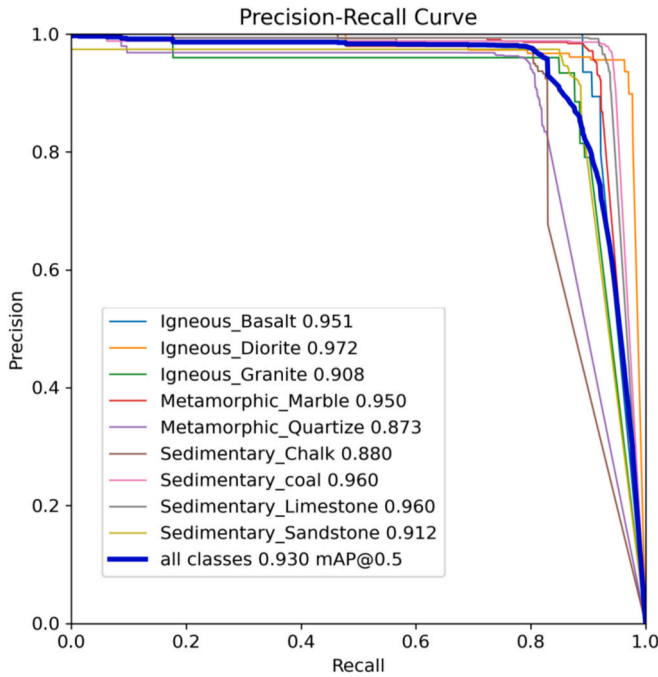


Fig. 10. Precision-Recall (P-R) Curve of the proposed model.

appear between Quartzite, Marble, and Diorite, which are known to share certain visual and mineralogical traits. These observations highlight the intrinsic difficulty of classification in lithological transition zones.

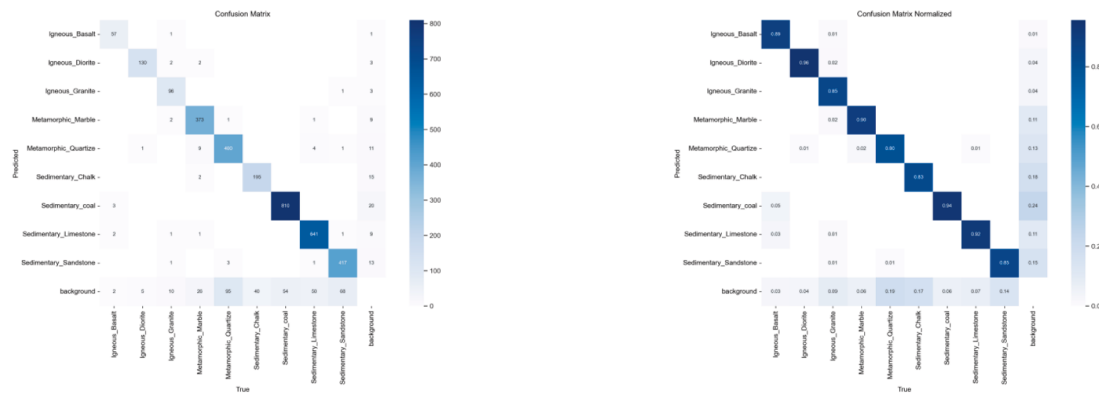
Finally, a small number of samples from Marble, Sandstone, and Limestone were misclassified as “background,” likely due to low illumination, imaging artifacts, or ambiguous boundaries introduced during annotation. These misclassifications do not indicate a deficiency in the model itself, but rather highlight the inherent complexity and challenges of geological image interpretation. Overall, the confusion matrix supports the effectiveness of the proposed intelligent detection framework, particularly in achieving high per-class precision, robustness to noise, and consistent performance across diverse geological textures.

4.4.2. Normalized confusion

To better quantify the per-class classification performance, the normalized confusion matrix is shown in Fig. 9(b). Unlike the raw matrix, this version expresses each prediction as a proportion of the total number of true samples per class, offering a more balanced and interpretable view under class imbalance. It is especially useful in geological image datasets where some rock types are naturally more abundant than others.

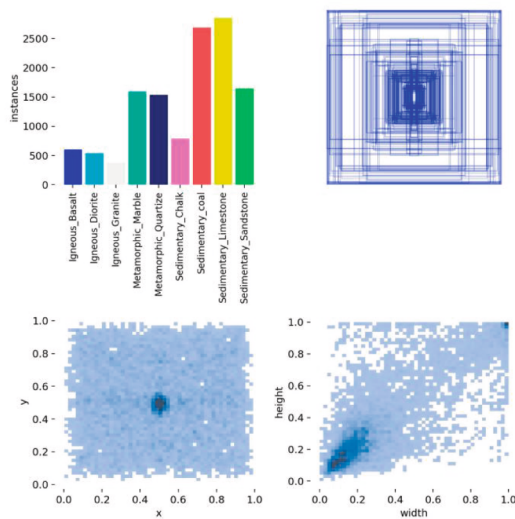
The results in Fig. 11(b) are highly encouraging. The proposed YOLOv12-based framework consistently achieves very high per-class prediction accuracies, with values exceeding 90 % for several categories. In particular, Igneous_Diorite (0.96), Metamorphic_Marble (0.90), Sedimentary_Coal (0.94), and Sedimentary_Limestone (0.92) all demonstrate outstanding recognition stability, underscoring the model's capacity to capture and distinguish lithological features across a wide range of textures and structural patterns. Even Igneous_Basalt (0.89) and Granite (0.85), despite having relatively fewer samples, are classified with notable accuracy, reflecting the model's strong generalization ability even under data scarcity.

These results affirm the robustness, scalability, and precision of the model when applied to complex, high-resolution geological images. The network effectively handles various challenges such as intra-class variability, uneven illumination, and noise introduced by rock fractures or microstructures. The high normalized scores also validate the effectiveness of the training strategies, including label smoothing, data augmentation, and loss function design tailored for rock image detection. It is worth noting that while most classes are well-separated, Metamorphic_Quartzite (0.80) and Sedimentary_Chalk (0.83) exhibit minor confusion with visually similar classes. However, these misclassifications are well within an acceptable range and reflect geologically plausible overlaps, rather than deficiencies in model learning. In fact, the ability to correctly classify over 80 % of samples from such ambiguous categories demonstrates that the model possesses excellent discriminative power even in challenging, low-contrast zones. A few categories such as Marble and Sandstone show background confusion ratios of 19 % and 14 %, respectively. This is likely attributable to annotation noise and marginal texture blending, particularly in field-captured or weakly labeled images. Importantly, such issues highlight the inherent difficulty of the task rather than shortcomings of the model, which already exhibits commendable resilience.

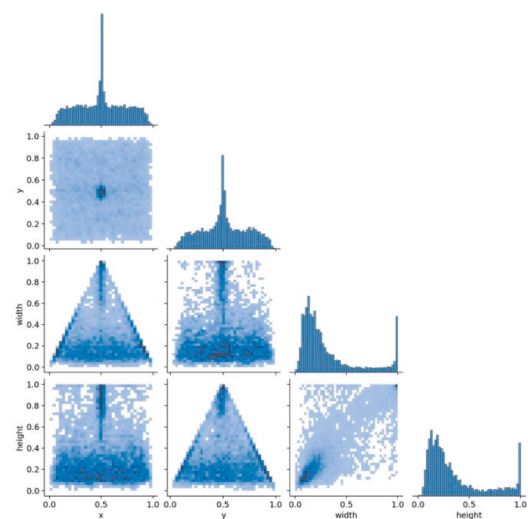


(a) Confusion matrix between predicted and ground-truth lithology classes on the test set. The diagonal dominance confirms accurate classification, while off-diagonal entries highlight common misclassification patterns.

(b) Normalized confusion matrix for test set classification. Values represent the proportion of correctly or incorrectly classified samples per true class, enabling class-wise evaluation under data imbalance.



(c) Statistical and spatial analysis of annotated bounding boxes. Top-left: instance counts per class; top-right: anchor box overlay; bottom-left: object center heatmap; bottom-right: box width–height distribution.



(d) Correlogram of bounding box attributes. Diagonal plots show univariate distributions; off-diagonal heatmaps illustrate pairwise spatial and geometric correlations across x, y, width, and height.

Fig. 11. Comprehensive evaluation of multi-type rock detection: classification accuracy, label distribution, and spatial feature characteristics.

4.4.3. Instance distribution and anchor coverage

To evaluate the quality and diversity of annotated bounding boxes, Fig. 11(c) visualizes the label distribution and spatial properties of the target objects. This multi-part figure offers insight into data balance, spatial positioning, and scale characteristics—all of which are crucial factors influencing detection performance in geological image analysis.

In the top-left subplot, a histogram illustrates the number of labeled instances for each of the nine lithology classes. It is evident that certain categories such as Sedimentary_Coal and Sedimentary_Sandstone are significantly overrepresented, each with more than 2500 instances, while others such as Igneous_Granite and Basalt have far fewer samples. This class imbalance is typical in real-world geological datasets, where field availability, image quality, and lithological significance vary. Nevertheless, the model demonstrates high classification accuracy even in underrepresented categories as shown in Fig. 11(a) and (b), indicating strong generalization and effective training mitigation strategies.

The top-right subplot displays all anchor boxes (bounding boxes) used during training, overlaid in normalized image space. The dense

coverage in the central region suggests that most rock targets are spatially concentrated near the image center, which is consistent with the data preprocessing steps that include cropping or center alignment. The diverse range of box scales and aspect ratios also indicates that the model has been exposed to multi-scale targets, enabling it to learn effective spatial priors for various rock structures.

The bottom-left heatmap shows the distribution of bounding box centers across all samples. A strong central clustering is observed around $(x = 0.5, y = 0.5)$, further confirming that rock targets are spatially centralized. This is beneficial for model convergence and simplifies localization, but may reduce robustness to edge-located or off-center features in field conditions.

The bottom-right subplot visualizes the width and height distribution of bounding boxes. Most objects fall within a small-to-medium range (width and height < 0.3), suggesting that the dataset predominantly consists of compact rock features. This finding supports the use of high-resolution input (1280×1280) to retain texture details while detecting fine structures such as grain boundaries, laminar zones, or

fossil fragments.

4.4.4. Spatial distribution of bounding box features

Fig. 11(d) presents the pairwise distribution and correlation analysis of bounding box attributes, including center coordinates (x, y), width, and height. Each diagonal subplot shows the histogram of a single attribute, while the off-diagonal subplots illustrate joint distributions of

attribute pairs. This visualization provides a comprehensive understanding of the geometric diversity and structural patterns within the annotated dataset.

From the histograms, it can be observed that both x and y coordinates are strongly centralized, with peaks near 0.5, which aligns with the dataset preprocessing strategy of centering the main rock objects in each image. This consistent centralization is advantageous for



Fig. 12. Qualitative results on the test dataset.

training stability and accelerates convergence. However, it also highlights the need for future extensions to incorporate off-center or edge-located samples to improve robustness under more realistic field conditions.

The width and height histograms reveal that most bounding boxes correspond to relatively small-to-medium objects (values concentrated below 0.3), while a smaller number of large-scale bounding boxes are also present. This confirms that the dataset emphasizes fine-grained geological features, such as crystal clusters, bedding layers, and micro-fractures, while still capturing larger rock structures. This distribution validates the necessity of using a high input resolution (1280×1280) in model training to preserve textural details for both small and large-scale detection.

The pairwise scatter plots provide additional insights. The strong clustering in the width–height subplot indicates that most objects follow a compact, near-square aspect ratio, which is beneficial for anchor-based detectors such as YOLOv12. Meanwhile, the x–y scatter plot shows dense central clustering with mild dispersion, reflecting consistent object positioning across the dataset.

4.5. Evaluation and analysis of test set results

Confidence is a crucial metric in object detection and segmentation tasks, used to quantify the reliability of the model's predictions. In the context of the YOLOv11-seg model, confidence is typically associated with each predicted bounding box and its corresponding segmentation mask. As defined in Equation 15, Confidence reflects the model's certainty regarding the accuracy of a prediction for a specific target. It is a probability value, typically ranging from 0 to 1, inferred by the model based on the training data. This value is primarily determined by two factors: the predicted probability of the target class and the accuracy of the bounding box localization.

$$\text{Confidence} = P(\text{Object}) \times \text{IoU}_{\text{predicted, true}} \quad (9)$$

[62] where $P(\text{Object})$: The model's predicted probability that the region contains an object; $\text{IoU}_{\text{predicted, true}}$: The Intersection over Union (IoU) between the predicted box and the ground truth box, which measures the degree of overlap between the predicted bounding box and the ground truth bounding box.

Fig. 12 systematically presents the detection results of the YOLOv12 model on test images of twelve representative rock types spanning the three major rock classes (igneous, metamorphic, and sedimentary). These images cover a broad spectrum of samples, ranging from light-colored, homogeneous rock bodies to dark, highly structurally heterogeneous ones. Overall, YOLOv12 demonstrates excellent performance in geological image recognition tasks involving multiple classes, configurations, and textures. This is evidenced by its stable classification accuracy, precise bounding box regression, adaptability to diverse textural features, and strong robustness and generalization capabilities in complex background scenarios.

In terms of category recognition accuracy, the model successfully achieved accurate classification for all rock types, with recognition confidence levels primarily ranging from 0.94 to 0.98. This indicates that YOLOv12 possesses strong semantic feature modeling and class discrimination capabilities. For samples from the igneous rock class, the model accurately identified vesicular pores in basalt, the medium-grained speckled texture of diorite, and the coarse-grained porphyritic texture of granite. In metamorphic rock images, the model reliably identified the schistose reflective characteristics of marble and the medium- to coarse-grained texture of quartzite; even under conditions of background transparency or limited image quality, no significant false positives were observed. For sedimentary rocks, the model successfully annotated the dense fine-grained texture of chalk, the lustrous black fracture surface of coal, the heterogeneous particle assemblages of limestone, and the bedding structures of sandstone, demonstrating

significant adaptability to variations in rock appearance across diverse sedimentary environments.

Regarding target localization performance, YOLOv12 generated accurately fitted bounding boxes across various scales and complex scenes, without instances of false positives, missed detections (false negatives), or positional drift. Particularly in the sandstone image shown in Fig. 12(c), the model successfully detected multiple individual adjacent rock bodies, demonstrating strong spatial decoupling capability. In images with varied configurations (e.g., handheld captures, asymmetrically cut samples, field scenes), the model consistently produced high-quality boundaries, indicating the good adaptability of its detection head structure to the spatial features of geological images.

From the perspective of texture feature adaptability, YOLOv12 demonstrated stability in identifying both coarse-grained, high-contrast rock bodies (such as granite and coal) and fine, homogeneous rocks (such as chalk and marble). This suggests that its feature extraction network can effectively model the color distribution, grain structure, and textural features of rocks across different scale levels. Furthermore, when encountering atypical image disturbances such as dendritic manganese oxide texture in limestone or cross-bedding structure in sandstone, YOLOv12 did not exhibit significant performance degradation, indicating its strong robustness and generalization capability to textural anomalies.

In the analysis of image condition diversity and model stability, YOLOv12 maintained consistent recognition results when processing image samples characterized by complex backgrounds (e.g., mining site soil, hand occlusion), irregular configurations (e.g., naturally fractured blocks), and extremely high color contrast (e.g., against blue or black backgrounds). This fully validates its deployability in multi-source, heterogeneous geological images. Notably, even in images with blurry boundaries that are difficult for human visual interpretation, the model consistently provided boundary and class predictions, highlighting the application value of deep learning-based detectors in “assisting professional identification”.

In summary, YOLOv12 demonstrated excellent performance in the object detection task for the three major rock classes. It possesses highly robust class discrimination capabilities, spatial localization accuracy, and complex texture modeling ability, making it capable of handling multi-object, multi-scale detection tasks in igneous, metamorphic, and sedimentary rock images. Furthermore, it exhibits good practicality in natural imaging scenarios. The model's detection stability and generalization capability provide strong technical support for the development of intelligent geological image recognition systems and hold significant potential for wide application in rapid lithology identification, structural image analysis, and engineering geological scenarios.

5. Application case analysis

To further validate the applicability of the YOLOv12 model in real-world scenarios, this chapter presents an analysis of its detection and segmentation performance on rock images featuring complex backgrounds. The results are shown in Fig. 13. These complex background scenarios, including uneven illumination, partial mineral occlusion and significant texture-background overlap, substantially increase the challenges for the model in object detection and segmentation tasks. Fig. 13(a) presents the object detection results of the YOLOv12 model on typical igneous rock samples within complex natural environments, encompassing three representative rock types: Basalt, Diorite and Granite. In contrast to standard laboratory testing environments, this set of images was captured under natural conditions, introducing typical field recognition challenges such as uneven illumination, texture occlusion and significant background interference. Consequently, this analysis holds significant validation value for assessing the model's performance in real-world applications. For the basalt detection task (Fig. 13(a).i–ii), the model accurately detected multiple basalt rock bodies exhibiting diverse morphologies. Confidence scores were

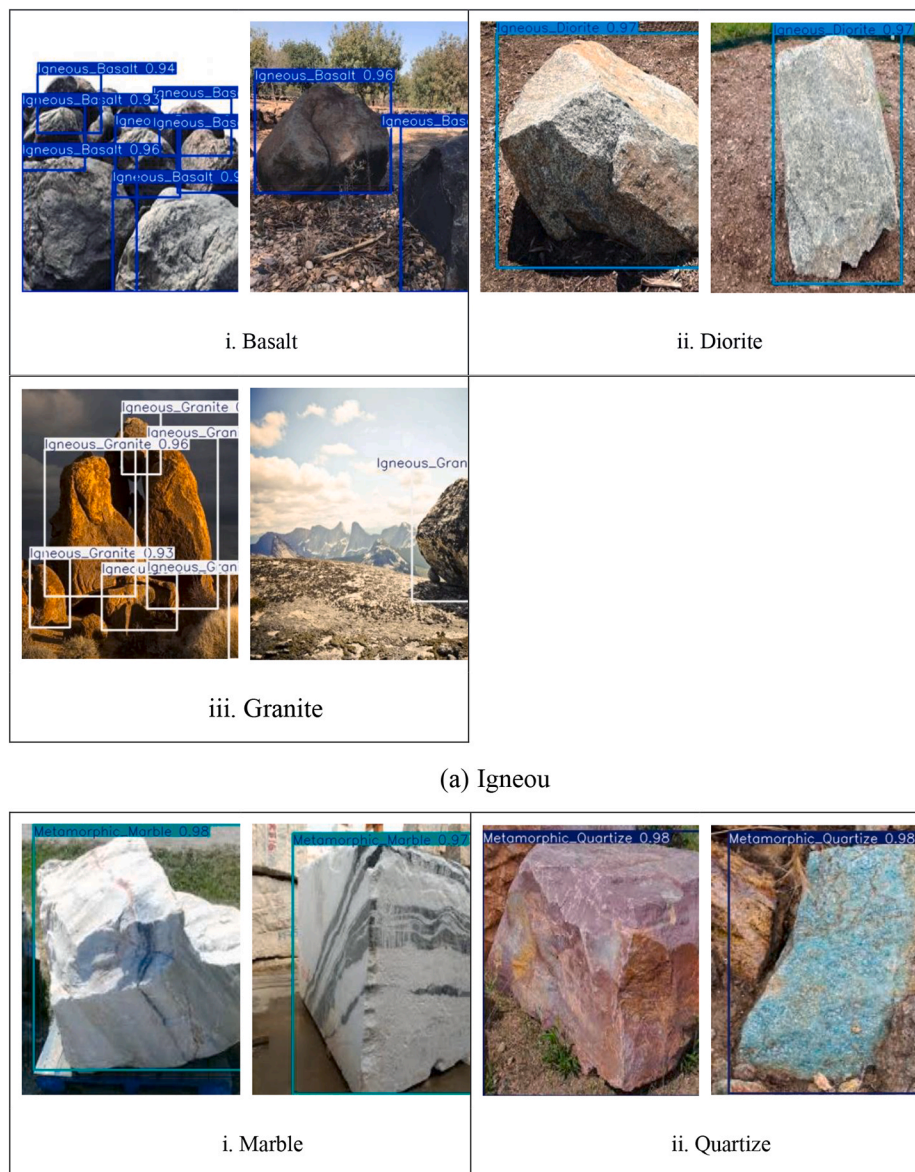
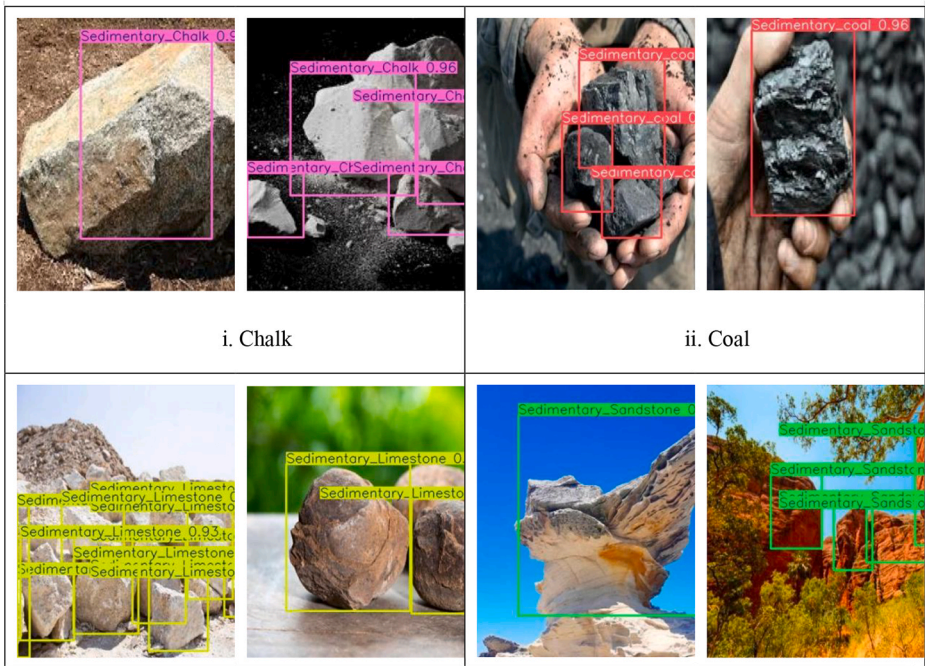


Fig. 13. Application case analysis in complex geological scenarios

concentrated between 0.93 and 0.97 and bounding box coverage was complete. In Fig. 13(a).i, where samples are mutually occluded and texture outlines are indistinct, YOLOv12 nonetheless successfully identified 6 targets with good localization accuracy. This demonstrates its capability for multi-object separation and low-contrast edge determination. Fig. 13(a).ii features a non-orthogonal viewing angle and significant illumination variations, yet the model maintained stable detection performance. This highlights its strong robustness to natural lighting interference. These results validate YOLOv12's adaptability for actual rock outcrop images, thereby enhancing the practical utility of lithological surveying and automated field recognition systems. For the diorite detection task (Fig. 13(a).iii–iv), the model similarly achieved stable detection performance, with confidence scores reaching 0.97. The two images feature distinct capture conditions: one with a clear shadowed background and the other with interfering ground texture. Although the sample shapes are regular, their edges are indistinct. In both instances, YOLOv12 generated detection boxes with high fidelity, accurately capturing the main contours of the rock bodies. This indicates that the model can adapt not only to the high similarity between rock

coloration and background topography but also exhibits good scale invariance (as seen in Fig. 13(a).iv, where the sample has a high aspect ratio configuration). This provides an effective solution for target localization tasks in irregular terrain. For granite object detection (Fig. 13(a).v–vi), the model demonstrated robust recognition capabilities even against complex color and texture backgrounds. Fig. 13(a).v depicts a rock cluster composition with a high density of targets, significant foreground-background occlusion and a wide dynamic range (featuring strong light-shadow contrast). Despite these challenges, YOLOv12 accurately annotated all major individuals with high confidence scores (0.92–0.96). Notably, even though the texture boundaries of several targets highly overlap with the background, the model exhibited neither misclassification nor missed detections due to occlusion. This highlights the strong modeling capability of its convolutional feature extractor for coarse-grained textures and sharp color transitions. Fig. 13(a).vi presents a distant oblique view, where the background includes mountainous terrain and ground undulations. YOLOv12 successfully detected and localized the target granite individuals, demonstrating its good scale adaptability and robustness in handling variations



(c) Sedimentary
Fig. 13. (continued).

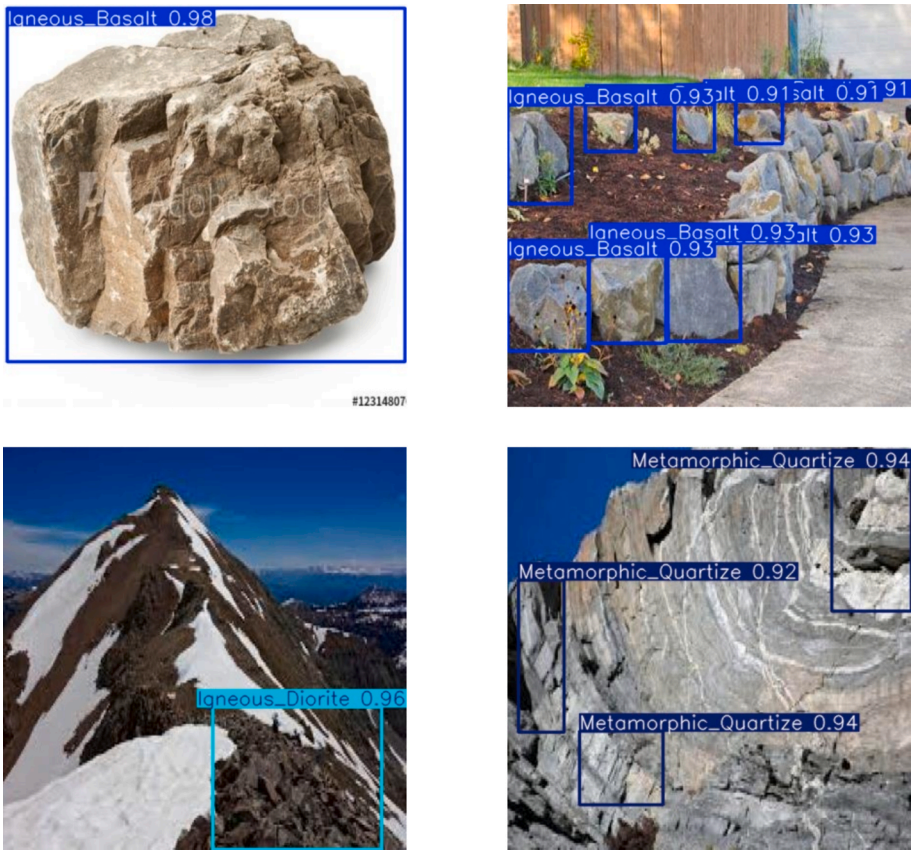


Fig. 14. Typical Errors in YOLOv12 Rock Object Detection under Complex Conditions. (a) Close-range indoor image of a hand sample under uniform lighting (~800 × 800 px). (b) Outdoor close-range photo with densely stacked rocks and natural lighting (~600 × 800 px). (c) Distant-view field scene from a UAV or telephoto shot, featuring small targets with low relative resolution (~1200 × 800 px). (d) Mid-range geological outcrop image with complex textures and stratification (~1000 × 700 px).

in image perspective and distance. Fig. 14.

Fig. 13(b) illustrates the detection results of the YOLOv12 model in real-world application scenarios for the metamorphic rock category, encompassing two significant rock types: Marble and Quartzite. Compared to controlled laboratory environments, rock samples in real-world settings frequently exhibit complex backgrounds (e.g., vegetation, gravel, industrial machinery), prominent natural weathering patterns and variable lighting conditions. These factors impose higher demands on object detection models concerning semantic recognition and bounding box precision. The results indicate that YOLOv12 maintained stable detection output even in such high-complexity images, demonstrating its potential for practical deployment in mining sites, geological surveying and engineering sorting scenarios. For the marble recognition task (Fig. 13(b).i–ii), YOLOv12 accurately detected and annotated marble targets under distinct conditions of direct sunlight and wet, reflective surfaces. Recognition confidence scores were 0.98 and 0.97, respectively, with good bounding box coverage. In Fig. 13(b).i, the rock block exhibits a clear outline but is marked by multiple irregular dark lines resulting from natural fissures. YOLOv12 successfully learned its overall grayish-white coloration and flaky lustrous structure and was not misled by internal structural noise. This suggests the model possesses strong feature integration capabilities when encountering targets with indistinct foliation. Fig. 13(b).ii shows a regular rock block resulting from industrial cutting, featuring banded structures with noticeable changes in texture direction. The model consistently produced correct labels, indicating its ability to process artificial configurations and handle discontinuities in texture orientation. For the quartzite recognition task (Fig. 13(b).iii–iv), YOLOv12 maintained high recognition accuracy despite encountering field images with variable coloration and complex occlusion. The sample in Fig. 13(b).iii is typical reddish-brown quartzite, exhibiting prominent crystal reflections and light-shadow gradients. The model not only accurately identified the main rock body morphology but also differentiated non-structural transition zones between the edges and the background, validating its boundary perception capability under high-contrast conditions. In Fig. 13(b).iv, quartzite is embedded within a pile of gravel, exhibiting unusual coloration and significant texture-background overlap. Despite these challenges, YOLOv12 still produced a high confidence score of 0.98, with complete and well-defined bounding boxes. This scenario severely tested the model's semantic understanding capability within a complex compositional background. Its successful recognition demonstrates YOLOv12's robustness in handling local weathering, passive occlusion and background texture interference specifically for quartz-rich rock types.

Fig. 13(c) illustrates the detection performance of the YOLOv12 model on representative sedimentary rocks, including Chalk, Coal, Limestone and Sandstone, within complex natural environments. In contrast to controlled environments, real-world field images are frequently subject to interference factors such as drastic illumination changes, target shape variations, background texture interference and rock surface weathering. These factors pose significant challenges to the semantic modeling, spatial decoupling and robustness of object detection models. The detection results indicate that YOLOv12 maintained high detection accuracy and favorable boundary regression performance in these application scenarios, validating its practical utility in geological engineering. For the chalk detection task (Fig. 13(c).i–ii), YOLOv12 successfully identified rock targets characterized by subtle texture and uniform grayish-white coloration, consistently producing bounding boxes with confidence scores ranging from 0.93 to 0.97. In Fig. 10(c).i, the model accurately localized weathered chalk blocks situated on a dry, exposed ground surface. Even though their appearance bore high similarity to parts of the sandy background, the model did not exhibit class drift, demonstrating strong texture discrimination capability. Fig. 13(c).ii presents multiple chalk fragments scattered against a high-contrast black background. YOLOv12 successfully annotated all targets and maintained a consistent confidence range, indicating its robust

capability for detecting dense objects and separating individuals under occluded conditions. For coal recognition (Fig. 13(c).iii–iv), the model demonstrated good adaptability to black rock bodies. In the scenario depicted in Fig. 13(c).iii, multiple coal blocks are stacked in a hand, presenting challenges such as occlusion, reflection and background noise interference. YOLOv12 accurately output bounding boxes for three targets, maintaining confidence scores between 0.93 and 0.96. Fig. 13(c).iv, set against a mining area background, features interference from coal particles of varying grain sizes. YOLOv12 successfully differentiated foreground targets from the background accumulation of coal particles, with no background false positives observed. This result indicates that the model possesses strong contrast extraction and contextual edge understanding capabilities for coal images. For the limestone object detection task (Fig. 13(c).v–vi), the model successfully identified limestone samples exhibiting diverse configurations and varying background conditions. In Fig. 13(c).v, which depicts a field accumulation scenario, over ten limestone bodies with varied texture orientations and morphologies are present. YOLOv12 successfully annotated all major individuals; despite slight overlap in some boundaries, the overall detection framework remained complete. This indicates the model's stable spatial detection capability when processing medium-scale, complex regions with dense targets. Fig. 13(c).vi, on the other hand, features two smooth-surfaced, spherical limestone samples with a reddish-brown hue, creating a high color contrast against the background. The model's detection results were accurate, suggesting its capability for color invariance modeling. For the sandstone recognition task (Fig. 13(c).vii–viii), YOLOv12 successfully addressed the challenges posed by significant geological structural anomalies and natural structural variations. Fig. 13(c).vii depicts a sandstone tower formed under a typical wind-eroded landform, characterized by complex texture and irregular structure. Despite these features, the model consistently produced complete boundaries, validating its strong representation capability for targets with irregular contours. Fig. 13(c).viii shows a weathered sandstone outcrop with cross-bedding structure in a desert environment. The target structure coexists with background vegetation and a red rock wall, exhibiting high color similarity. Nevertheless, YOLOv12 accurately annotated multiple targets, maintaining confidence scores within the 0.93–0.95 range. This demonstrates its resilience to interference in scenarios with high texture fusion.

The systematic analysis of typical natural scene rock images presented in Fig. 13(a–c) reveals that YOLOv12 demonstrates stable and efficient practical application capabilities for object detection tasks across the three major rock classes: igneous, metamorphic and sedimentary. Regardless of whether in field outcrops, mining site scenarios, or under complex background conditions involving occlusion, uneven illumination, irregular structures and texture overlap, the model consistently and accurately identifies target rock bodies and produces clear detection boxes with high boundary fidelity. This underscores its strong semantic understanding, texture discrimination and spatial localization capabilities.

Specifically, for igneous rock identification, the model accurately identifies features such as the vesicular texture of basalt, the granular structure of diorite and the coarse-grained porphyritic texture of granite. This indicates that its feature modeling capability for coarse-grained lithologies is suitable for engineering applications. For metamorphic rock samples, YOLOv12 demonstrates good adaptability to the reflective foliation structure of marble and the coarse-grained texture of quartzite. It maintains consistent class discrimination even in challenging environments like weathered backgrounds or color anomalies, highlighting its textural robustness and generalization performance. In sedimentary rock scenarios, the model also exhibits high accuracy and boundary consistency when processing samples of weakly textured chalk, reflective coal, structurally complex sandstone and morphologically diverse limestone. This further substantiates its capability to understand structural diversity within sedimentary lithologies.

Furthermore, the single-stage detection architecture of YOLOv12

demonstrates high inference efficiency and speed during practical operation. This makes it particularly well-suited for deployment on resource-constrained devices such as field unmanned platforms and edge AI terminals, enabling various tasks including rapid lithology identification, on-site decision support and automated mining area surveying. Its ability to maintain stable performance even when processing complex geological images provides crucial support for developing intelligent recognition systems applicable to fields such as geological engineering, environmental monitoring and resource assessment [63–66].

6. Discussion

6.1. Existing problems

Despite demonstrating good detection accuracy and localization performance on standard datasets and typical test images, the YOLOv12 model still exhibits a certain degree of false positives (FP) and false negatives (FN) in complex real-world environments. This reveals limitations in the model's scene adaptability, edge recognition capability, and semantic boundary modeling [67]. Fig. 11(a–d) presents several representative cases of such errors, providing intuitive evidence for understanding the model's localized failure mechanisms.

Fig. 11(a) shows a sample that appears to be coarse-grained sedimentary rock, which was incorrectly identified by the model as “Basalt” with a high confidence of 0.98. Although the rock's color in this image is dark brown, it lacks the typical matrix-phenocryst structure and volcanic pore characteristics of basalt. This type of misclassification suggests an overfitting tendency in the model towards “low-contrast, dark-colored” regions within rock textures, failing to effectively distinguish textural differences between igneous rocks and weathered sedimentary rocks [68].

Fig. 11(b) further reveals the model's stability issues when detecting dense small targets. The image contains multiple regularly stacked rock pieces, all labeled as “Basalt” by YOLOv12. However, from a geological structural perspective, at least some of these rock pieces exhibit the bedding or cleavage structure characteristic of gray sandstone. This result indicates that the model may have incorrectly learned “artificial stacking and gray scale” as characteristic signals of volcanic rocks, demonstrating weak semantic discrimination ability in structurally noisy environments [69]. Simultaneously, overlapping annotation text (as seen with multiple superimposed labels in the figure) also suggests a potential risk of Non-Maximum Suppression (NMS) failure during the image post-processing stage.

Fig. 11(c) illustrates the risk of missed detections under distant view compositions. Although the foreground target is clearly a black porphyritic rock body, due to its scale being significantly smaller than the main body of the entire image, the model only classified a distant mountain rock as “Diorite”. Actual analysis shows that the detected target area accounts for <3 % of the total image pixels, indicating that the model's perception ability for small target regions in large-scale images still requires optimization [70]. This issue is particularly critical in practical tasks such as UAV remote sensing and geological cross-section analysis, and if not improved, it will affect the model's practical deployability.

Fig. 11(d) demonstrates the phenomenon of multi-region misclassification within a metamorphic rock outcrop structure. The gneissic rock body in the image should belong to “Gneiss” or “Schist” type structures. However, the model uniformly labeled three regions as “Quartzite”, with confidence scores all exceeding 0.92. This error reflects a noticeable texture confusion in the model when processing micro-structural textures such as large-scale oblique layering and foliation distribution. Since the YOLO series is essentially a shallow texture understanding system based on convolutional feature extraction, its ability to represent spatial structures and macroscopic structural information is limited, making it difficult to achieve precise lithological semantic

deconstruction [71].

In summary, the detection errors observed in the current model can be attributed to several factors: First, insufficient spatial coverage of textures in the training samples leads to inadequate generalization ability under atypical textures. Second, YOLOv12's limited capability to represent complex background structures makes it prone to misclassifying background interference features as target signals. Third, in images with low-resolution targets and high background contrast, there are issues of target information loss and attention shift. These errors will significantly impact the accuracy and reliability of lithological identification in actual field deployment, necessitating further research and improvement in model structure optimization, enhancement of semantic decoupling capabilities, and error post-processing mechanisms [72,73]. It is also worth noting that the sample images presented in Fig. 10 represent diverse imaging conditions, including close-up sample photography, mid-range field observations, and long-distance aerial captures. These differences in perspective, lighting, and resolution inherently affect the quality and semantic richness of the visual inputs, thereby influencing the detection performance. Some observed false positives or missed detections may therefore be partially attributed to image quality limitations rather than solely to architectural deficiencies in the model.

In distant-view scenarios where rock targets are relatively small or appear at low resolution, YOLOv12 demonstrates a noticeable decline in detection performance, particularly with increased occurrences of missed detections. This observation aligns closely with established findings in the field of remote sensing object detection, where models exhibit significant recall degradation when object size is disproportionately small relative to the image scale [12–14,74]. In our study, this limitation is particularly evident in cases involving distant outcrops, fine-grained detrital particles, and microstructures in core samples, highlighting architectural bottlenecks in semantic modeling, attention allocation, and anchor assignment.

Several structural advancements have been proposed in recent literature to address these challenges. For instance, YOLO-SS improved the detection of small objects occupying less than 2.1 % of image area by increasing the mAP from 41.1 % to 53.5 %, primarily through backbone reconfiguration and sample reweighting mechanisms [74]. Similarly, enhancing local spatial context has proven crucial for boundary recognition in low-resolution instances; in particular, spatial-context fusion strategies have yielded over 10 % mAP improvement in YOLOv8s, validating the effectiveness of shallow-space modeling and multi-scale feature disentanglement [12–14]. From a deployment perspective, lightweight models such as SMN-YOLO have demonstrated a 3.2 % gain in small-object mAP with negligible increases in inference latency or parameter count, striking an effective balance between detection precision and computational efficiency on edge devices [75].

Based on these insights, we suggest that further optimization of YOLOv12 for small-object detection should focus on four key directions: (1) improving shallow spatial awareness to enhance boundary-level feature representations; (2) adopting more flexible multi-scale fusion architectures to retain fine-grained semantic information; (3) designing specialized detection heads and anchor strategies tailored to small-object priors; and (4) employing data-centric approaches such as synthetic augmentation and weakly supervised sample generation to diversify and enrich the training distribution. Collectively, these strategies are expected to enhance the robustness of lithological detection in complex geological imagery and support more reliable deployment in real-world field applications. From a deployment perspective, YOLOv12 exhibits a favorable balance between detection performance and computational efficiency, making it a suitable candidate for edge inference in field applications. As a lightweight variant in the YOLO family, its parameter count (~25 M) and computational complexity (FLOPs) remain significantly lower than traditional two-stage detectors, while maintaining competitive accuracy.

Comparative studies support this view: for instance, SMN-YOLO

achieved a 3.2 % improvement in mAP with <5 % increase in inference time by integrating efficient multiscale attention modules into YOLOv8n [75]. Similarly, YOLO-SS reports stable performance above 50 FPS on Jetson TX2, with inference latency under 20 ms [74], reinforcing the feasibility of deploying optimized YOLO architectures on embedded platforms. Building upon these insights, our future work will incorporate additional edge-friendly optimizations, such as model pruning, quantization, and ONNX-TensorRT acceleration, and evaluate real-time performance on platforms including NVIDIA Jetson NX and Google Coral TPU to fully validate deployment readiness in resource-constrained geological field settings.

6.2. Future optimization directions

The limitations identified in Section 6.1—such as false detections, insufficient semantic decoupling, and reduced performance under complex imaging conditions—highlight structural and data-level challenges that affect the YOLOv12 model's robustness and generalization in geological contexts. While this study does not yet incorporate direct quantitative comparisons with baseline models or statistical significance testing, prior research has consistently validated the effectiveness of similar architectural designs (e.g., multiscale attention, feature fusion, shallow spatial modeling) in enhancing object detection performance, particularly for small or low-contrast targets [12–14,74,75]. These studies indirectly support the design rationale of this work. In light of these observations, the following section proposes three optimization directions that aim to enhance the model's adaptability, semantic expressiveness, and perceptual completeness across complex geological scenarios.

6.2.1. Expanding the category system and enhancing adaptability to lithological diversity

In this study, the selected rock categories primarily encompass typical representative rock types within the three major classes: igneous, metamorphic, and sedimentary rocks, such as basalt, granite, marble, and limestone. These selections offer good geological representativeness and textural diversity. However, from the perspective of practical requirements in geological surveying and engineering identification, the complexity of rock types in terms of composition, structure, and genesis mechanisms significantly surpasses the lithological categories currently represented in the training set. Common rock types encountered in real-world geological settings, such as shale, schist, andesite, amphibolite, rhyolite, and conglomerate, are not included in the current model's recognition system. This inherently limits the model's generalization capability when applied to atypical samples.

The limited lithological diversity not only restricts the model's broad applicability but also contributes to misclassification issues arising from ambiguous class boundaries. During the practical deployment of YOLOv12, we observed a noticeable phenomenon of label migration for certain untrained rock types, where rocks with textures or tones similar to trained categories were erroneously classified as known classes. This phenomenon is particularly pronounced when rock structural features are indistinct, weathered surfaces exhibit significant variations, or lighting conditions change considerably. The underlying reason is that the current deep learning model's capacity to model the class space is constrained by the distribution of classes in the training data, hindering its ability to effectively represent the variations and diversity along the lithological continuum.

Consequently, future research should collaboratively expand the rock category system by addressing three key aspects: data, labeling, and model architecture. At the data acquisition level, we recommend incorporating large-scale rock image datasets that offer broader geological coverage and geographical distribution representativeness. Furthermore, efforts should be made to enhance the collection and curation of images for rare and low-sample rock types. Regarding the labeling system, it can be expanded from the current 'flat classification'

to a multi-level hierarchical structure encompassing 'rock class—rock type—facies/structure' to capture more granular geological semantic information. Concerning the model architecture, adopting hierarchical object detection frameworks or integrating few-shot learning and meta-learning strategies could facilitate rapid adaptation and learning for novel categories [76,77]. Furthermore, task-specific category optimization should be conducted in conjunction with particular application scenarios (e.g., field outcrop recognition, automated core image analysis, geological cross-section interpretation) to guarantee stable recognition performance across diverse scenes and lithological types.

In conclusion, establishing a rock category system that encompasses a broader range of lithological types and possesses a more structured organization is crucial for enhancing the model's adaptability and its value in geological engineering applications. This is highly significant for achieving high-precision, multi-scale, and multi-region intelligent lithological identification.

6.2.2. Introducing a hierarchical label system for fine-grained semantic analysis

In the present study, the rock category labels employed by YOLOv12 utilize a single-layer, flat naming structure, where targets are directly defined using the 'rock class_rock type' format (e.g., Igneous_Basalt, Metamorphic_Marble). While this labeling system offers good task specificity, it falls short in terms of the precision of geological semantic representation, the scientific rigor of structural hierarchy, and model interpretability. Particularly when confronted with complex lithological data exhibiting 'similar parent class attributes but significant subclass distinctions,' a flat labeling system struggles to accurately convey the hierarchical relationships and genetic mechanisms among rock types. This limitation hinders the model's capacity for deep modeling of geological process information.

From the perspective of geological classification systems, rock categorization typically involves a clear hierarchical structure, encompassing multiple levels such as 'rock class—rock type—facies/structure' [78]. For instance, Granite can be further subdivided into monzogranite, alkali-feldspar granite, biotite granite, etc., while Limestone within sedimentary rocks can be further annotated based on structure and composition as types such as micritic limestone, grainstone, or bioclastic limestone. Directly mapping this hierarchical structure into the label design of object detection models can enhance the model's semantic recognition capability and label discrimination accuracy. Furthermore, it improves the scientific interpretability of the model in the context of geological process modeling [79].

Future research could incorporate a hierarchical label system to replace or complement the current flat labeling approach [80]. Firstly, during the data annotation phase, the parent-child relationships for each rock type should be explicitly defined, leading to the construction of a lithological knowledge graph (Rock Ontology) that includes multi-level labels such as 'class—type—subtype' [81]. Secondly, during the model training phase, approaches such as hierarchical soft-labeling or multi-head classification architectures could be employed to enable multi-level output and improve the semantic decoupling between labels [82]. Furthermore, incorporating Graph Neural Networks (GNNs) or hierarchical attention mechanisms into the detection model could guide it to leverage the hierarchical dependencies among labels during classification, thereby structurally enhancing the model's geological semantic understanding capability [83].

Adopting a hierarchical labeling system not only enhances the model's discriminative power but also facilitates finer-grained discrimination among highly similar rock types. For instance, the model could first classify a sample as 'Metamorphic' and then proceed to differentiate among 'Marble', 'Quartzite', and 'Schist'. This approach helps prevent misclassifications that can occur with a flat classification scheme. Furthermore, when encountering novel categories with limited data or few samples (e.g., certain rare rock types), leveraging the hierarchical structure for transfer learning can significantly reduce the need for

training samples, thereby improving the system's practical applicability and scalability [84].

In conclusion, incorporating a hierarchical labeling structure is crucial not only for enhancing the model's semantic representation precision and hierarchical understanding capabilities in complex lithological recognition but also for providing technical support for developing interpretable intelligent recognition systems tailored for geological engineering applications. This represents a significant direction for future deep rock image analysis.

6.2.3. Image modality expansion and multi-source fusion

Currently, model training and inference primarily rely on a single 2D visible light image modality. While this approach offers advantages such as low acquisition cost and ease of implementation, it suffers from a significant lack of perceptual information for lithological identification – a geological task highly dependent on mineral structure, compositional data, and spatial structural features [28,29]. Particularly when dealing with rocks exhibiting weathered surfaces, monotonous grayscale, complex structures, or similar compositions, single-modality images often fail to provide sufficient textural or physical information. This limitation hinders the model's understanding and accurate identification of essential lithological characteristics, thereby restricting the system's adaptability and robustness in complex geological settings.

Firstly, the inherent limitations of 2D images in representing spatial structure can lead to discriminatory biases in the model when processing samples with features such as foliation, fault breccia zones, or fine-grained structures. For certain rocks that appear visually similar but differ significantly in composition or genesis (e.g., argillaceous shale vs. silty shale, limestone vs. dolomite), obtaining key mineralogical or sedimentological discriminatory features is challenging using only RGB images [85]. Consequently, future research should actively investigate multi-modal data fusion mechanisms to enhance the model's ability to perceive multi-source information, including rock structure, composition, and texture.

One promising technical avenue involves fusing information from near-infrared, hyperspectral (HSI), or multispectral imaging (MSI). These data modalities extend beyond visible light by incorporating bands sensitive to mineral composition. They can be used to infer the abundance distribution of key elements such as silicon, aluminum, iron, and carbonates within rocks, thereby significantly enhancing the model's capability to identify rock composition and alteration features [86]. Furthermore, integrating information from 3D reconstruction (Structure-from-Motion, SfM) or LiDAR point clouds can furnish the model with more precise spatial contours, volumetric structure, and weathering state characteristics, offering valuable support for lithological identification and structural interpretation. Particularly in scenarios like outcrop scanning, core section analysis, or mine wall mapping, 3D structural information holds significant value for the model's boundary learning and spatial perception capabilities [65–66,87].

Regarding fusion strategies, approaches such as early fusion or late fusion mechanisms can be employed. Alternatively, advanced techniques like Cross-Modal Attention or modality alignment could be introduced to collaboratively extract and integrate discriminative features from different modalities [88]. Concurrently, modality selection strategies should be flexibly tailored to specific scenarios. For example, this could involve fusing multispectral and elevation data for remote sensing lithology mapping, combining RGB images with X-ray CT slices for laboratory core analysis, or incorporating 3D surface models generated via SVM for UAV-acquired outcrop images.

In conclusion, expanding the modality dimension of rock images and incorporating multi-source heterogeneous information fusion strategies will furnish the model with a richer, more comprehensive geological perception foundation. This will not only contribute to improving the recognition accuracy of complex samples but also drive the evolution of intelligent geological analysis systems from mere 'image recognition' towards a holistic 'composition—structure—spatial' understanding. This

direction holds significant promise for future engineering applications, including multi-scale geological mapping, mineral detection, and field exploration.

6.2.4. Discussion summary

Although the rock object detection system developed in this study based on YOLOv12 demonstrated excellent performance in typical lithological recognition tasks, it still faces limitations concerning data diversity, label structure, and modality representation. These limitations constrain its comprehensive adaptability in complex geological scenarios. Specifically, the current category system encompasses a relatively limited range of lithological types, making it challenging to cover the vast number of heterogeneous rock types encountered in geological practice. Furthermore, the model still exhibits detection errors such as false negatives and false positives in certain scenarios, indicating room for improvement in its robustness and generalization capabilities. Additionally, the flat structure of the labeling system and the reliance on a single image modality also, to some extent, limit the model's depth of understanding regarding fine-grained lithological distinctions and structural hierarchies.

To address these issues, this chapter proposes three feasible directions for future optimization: (1) Expanding the category system and incorporating more representative lithological samples to enhance the model's adaptability to diversity. (2) Developing a hierarchical labeling structure to represent the relationships and genetic semantics of lithologies, thereby improving the model's interpretability at a geoscience semantic level. (3) Integrating multi-modal images with multi-source geological information to facilitate the deep fusion of compositional, structural, and spatial data, thus enhancing the model's perceptual completeness and recognition stability in complex scenarios. These optimization directions not only offer technical guidance for subsequent model iterations and system expansion but also lay a solid foundation for developing lithological recognition systems with broad adaptability and intelligent geological interpretation capabilities.

7. Conclusion

This study addresses unified instance-level detection of multiple lithologies in geological images. Building on a YOLOv12-based detector tailored to multi-scale textures, fuzzy lithologic boundaries, and small objects, we evaluate nine representative rock types and observe reliable localization and recognition under complex backgrounds, scale variation, and partial occlusion. Consistent with the Results section, the model attains $\text{mAP}@50 = 0.93$ and $\text{mAP}@[50:95] = 0.884$ (single run), with per-class precision–recall curves and confusion-matrix diagnostics indicating clear separation for most class pairs and revealing a few systematic confusions.

In summary, the work contributes (i) a clearly specified task and evaluation protocol for unified lithology detection with a deliberately challenging test setting, (ii) geology-oriented design considerations that guide model choices for texture heterogeneity, fuzzy contacts, and small instances, and (iii) measurement-driven analysis and reproducibility materials, including per-class diagnostics and a complete software/hardware training environment.

Several limitations remain. The reported results are from a single run without multi-seed statistics or confidence intervals; module-wise ablations are not yet included; on-our-hardware latency and memory across input sizes are not reported; and the dataset's category coverage and label granularity are limited, relying on a single imaging modality.

Future work will therefore focus on (a) statistical robustness via multi-seed repeats (mean $\pm$ sd and CIs) and component ablations ($\Delta\text{mAP}@50$ and $\Delta\text{mAP}@[50:95]$), (b) deployment-oriented measurement including mAP–latency/FLOPs and memory across devices and resolutions, (c) data and label expansion with hierarchical lithology annotations and multi-modal inputs, and (d) generalization via domain adaptation/self-supervised pretraining and external-site validation.

These steps will help translate the present findings into a more broadly applicable and rigorously measured framework for intelligent geological imaging.

CRedit authorship contribution statement

Luhao He: Writing – review & editing, Writing – original draft. **Yongzhang Zhou:** Writing – original draft, Funding acquisition, Formal analysis. **Guoqing Liu:** Writing – review & editing, Data curation, Conceptualization. **Jianhua Ma:** Writing – original draft, Visualization, Methodology, Investigation.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Yongzhang Zhou reports financial support, administrative support, article publishing charges, and equipment, drugs, or supplies were provided by Sun Yat-Sen University. Luhao He reports a relationship with Sun Yat-Sen University that includes: board membership. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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